# Online Resource 2 for "Machine-read settlement names did not detectably improve 250 m predictions of mapped river flooding, in a test insensitive to effects near settlements": archive of further tables

This file keeps the tables of the exploratory analyses that neither the paper nor Online Resource 1 cites (the account of the earlier test in Japan is in Text S1 of Online Resource 1), including tables whose quantities were later corrected or superseded; each caption names the amendment that produced it, and the amendments in the public repository record the corrections. Nothing here is needed to read the paper, and no finding rests on these tables alone. Tables numbered S$k$ are in Online Resource 1. Everything outside the frozen protocol is exploratory (Table S5 of Online Resource 1 gives the planning status of each group of estimates).

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# Pre-specified analyses

Table A1: Secondary comparisons under each split: the total increment of names over the recalibrated baseline model (including the regional composition of names) and the flexible model refitted with the name features, for the lexicon and learned score and, on the language-model split, for the language-model reading. The 95 % interval is the fixed-margin interval of the decision table (Sect. 2 of the paper); the fourth column gives the interval from resampling the ratio of the two increments, and the fifth the interval of $R$ minus the same comparison with names permuted among the settlements of each AOI.

| Split: comparison | $R$ | 95 % interval | Ratio bootstrap, 95 % | real − placebo (95 %) | Verdict | TOST |
|---|---|---|---|---|---|---|
| Location groups: total, M1s→M3s | +0.031 | [−0.051, +0.155] | [−0.054, +0.192] | [−0.015, +0.030] | inconclusive | no |
| Location groups: flexible, M1→M3 | +0.038 | [−0.058, +0.142] | [−0.072, +0.128] | [−0.003, +0.075] | inconclusive | no |
| Countries: total, M1s→M3s | −0.037 | [−0.149, +0.043] | [−0.275, +0.089] | [−0.052, +0.074] | inconclusive | no |
| Countries: flexible, M1→M3 | +0.040 | [−0.047, +0.158] | [−0.079, +0.391] | [−0.143, +0.077] | inconclusive | no |
| Time: total, M1s→M3s | +0.007 | [−0.101, +0.113] | [−0.329, +0.328] | [−0.149, +0.111] | undefined-denominator | — |
| Time: flexible, M1→M3 | −0.128 | [−0.513, +0.215] | [−2.026, +1.544] | [−0.168, +0.636] | undefined-denominator | — |
| Language-model cut-off: total, M1s→M3s | +0.001 | [−0.025, +0.028] | [−0.026, +0.087] | [−0.065, +0.022] | equivalent-null | yes |
| Language-model cut-off: flexible, M1→M3 | +0.168 | [+0.039, +0.328] | [+0.040, +1.126] | [−0.0004, +0.068] | not-meaning | no |
| Language-model cut-off: language model, total, M1s→M3rs | +0.027 | [+0.0003, +0.056] | [−0.001, +0.195] | [+0.007, +0.045] | positive-small | no |
| Language-model cut-off: language model, flexible, M1→M3r | +0.052 | [−0.023, +0.138] | [−0.034, +0.380] | [−0.161, +0.014] | inconclusive | no |

Table A2: Ratio-bootstrap 95 % intervals of $R$ for every comparison and split (post hoc, planned in amendment 2): the two increments are resampled together (cluster bootstrap over location groups, countries for the country hold-out, 4,000 replicates) and their ratio taken in each replicate. The decision table uses fixed-margin intervals instead (Sect. 2 of the paper), because the ratio mixes the uncertainty of the denominator into $R$. [u]The pre-specified rule leaves $R$ undefined on this split. —: comparison not run on this split.

| Comparison | Location groups | Countries | Time[u] | LM cut-off | Events grouped |
|---|---|---|---|---|---|
| Names within areas, M1c→M3n (H2) | [−0.016, +0.004] | [−0.008, +0.025] | [−0.089, +0.069] | [−0.014, +0.004] | [+0.001, +0.029] |
| Names permuted, M1c→M3np | [−0.035, +0.003] | [−0.046, +0.006] | [−0.297, +0.186] | [−0.034, +0.028] | [−0.015, +0.088] |
| Lexicon only, M1c→M3nx | [−0.019, +0.002] | [−0.002, +0.038] | [−0.113, +0.063] | [−0.015, +0.004] | — |
| Learned score only, M1c→M3nl | [−0.006, +0.001] | [−0.004, +0.028] | [−0.029, +0.039] | [−0.008, +0.002] | — |
| Position only, M0c→M0n (H1) | [−0.006, +0.002] | [−0.006, +0.020] | [−0.025, +0.025] | [−0.006, +0.008] | — |
| Position only, flexible, M0→M0N | [+0.006, +0.252] | [+0.042, +0.287] | [−7.715, +5.075] | [−0.187, +0.011] | — |
| Total, M1s→M3s | [−0.054, +0.192] | [−0.275, +0.089] | [−0.329, +0.328] | [−0.026, +0.087] | [−0.144, +0.093] |
| Flexible, M1→M3 | [−0.072, +0.128] | [−0.079, +0.391] | [−2.026, +1.544] | [+0.040, +1.126] | — |
| Language model, M1rc→M3rn (H4) | — | — | — | [−0.034, +0.013] | — |
| Language model, permuted | — | — | — | [−0.028, +0.003] | — |
| Language model, total, M1s→M3rs | — | — | — | [−0.001, +0.195] | — |
| Language model, flexible | — | — | — | [−0.034, +0.380] | — |
| Names centred within AOIs, M1s→M3d | [−0.012, −0.0004] | [−0.014, +0.0001] | [−0.041, +0.028] | [−0.005, +0.009] | [−0.005, −0.001] |

Table A3: Operating characteristics of the decision rule on synthetic labels for each split (the primary split before the freeze; the others post hoc, amendment 2), with 95 % Wilson intervals for the Monte-Carlo error. The simulated effect is $\delta$ times the indicator that the nearest name is water-related in the lexicon, calibrated so that $R \approx 0.05$ on the primary split ($\delta = 1.2$, an odds ratio of 3.3); on the other splits the same $\delta$ gives the mean $R$ shown, not the smallest effect of interest. Equivalent counts the replicates whose 90 % interval of $R$ lies within $\pm 0.05$, whatever their label. The last column gives, for $\delta = 1.2$, the share of replicates whose 90 % interval of $\Delta$BS lies within the equivalence margin of the real labels on the same split (post hoc, amendment 2): how often an effect of this size would have been called equivalent in our data (—: margin undefined on the time split). Margin ratio and upper ratio (amendment 2, addendum 3): the real-label equivalence margin and the upper end of the observed 95 % interval of $\Delta$BS on the same split, each divided by the mean $\Delta$BS that $\delta = 1.2$ produced there; these ratios ignore that intervals widen under the alternative, so the column before them is the better guide. [n]Includes replicates whose simulated baseline gain was negative (the interval of $R$ is then reversed; Appendix 2.4 of the paper). [a]The effect is injected through the lexicon, so this row is the lexicon test on the language-model split; there is no power analysis for the language-model reading (H4).

| Split | $\delta$ | Replicates | Equivalent (%) | Positive (%) | Mean $R$ | Labels | Equivalent, real margin (%) | Margin ratio | Upper ratio |
|---|---|---|---|---|---|---|---|---|---|
| Location groups (pre-specified) | 0 | 100 | 99 [95, 100] | 0 [0, 4] | −0.001 | equivalent-null 91, inconclusive 1, negative 7, not-meaning 1 | — | — | — |
| | 1.2 | 100 | 0 [0, 4] | 98 [93, 99] | +0.057 | inconclusive 1, positive-meaningful 69, positive-small 29, undefined-denominator 1 | 26 [18, 35] | 1.5 | 0.11 |
| Countries | 0 | 20 | 95 [76, 99] | 0 [0, 16] | −0.004 | equivalent-null 16, inconclusive 1, negative 3 | — | — | — |
| | 1.2 | 20 | 0 [0, 16] | 65 [43, 82] | +0.081 | not-meaning 4, positive-meaningful 12, positive-small 1, undefined-denominator 3 | 0 [0, 16] | 1.1 | 0.25 |
| Time | 0 | 20 | 100 [84, 100] | 0 [0, 16] | +0.004 | equivalent-null 12, negative 1, undefined-denominator 7 | — | — | — |
| | 1.2 | 20 | 65 [43, 82] | 5 [1, 24] | +0.017 | equivalent-null 10, inconclusive 2, positive-negligible 1, undefined-denominator 7 | — | — | 0.99 |
| Language-model cut-off, lexicon effect[a] | 0 | 20 | 100 [84, 100] | 0 [0, 16] | +0.0002 | equivalent-null 18, negative 1, undefined-denominator 1 | — | — | — |
| | 1.2 | 20 | 85 [64, 95] | 15 [5, 36] | +0.013 | equivalent-null 13, inconclusive 2, positive-negligible 3, undefined-denominator 2 | 100 [84, 100][n] | 4.2 | 0.18 |

Table A4: Descriptive strata for the primary comparison: the H5 strata of the protocol (first block, no intervals as pre-specified; built-up cover and continent are also strata of the protocol and are shown with post hoc intervals) and the post hoc strata by the first language of the country of the nearest settlement (languages with at least 20,000 cells), with 90 % intervals from the cluster bootstrap over location groups within the stratum. —: the baseline gain within the stratum is not positive, so $R$ is undefined. $R$ within a stratum is the names increment in the stratum divided by the baseline gain in the same stratum; small strata have small and unstable baseline gains, hence wide intervals. Language strata are the first language of the country in GeoNames (for example ur for Pakistan); in parentheses, the blocks of the lexicon that read the names of the analysed countries of the stratum (fa and hi for Pakistan). Strata with few location groups have intervals from few resamples, and the fa stratum, three location groups with a large baseline gain of their own, has a small $R$ and a narrow interval.

| Stratum | Cells | Flooded (%) | $R$ within stratum | 90 % interval |
|---|---|---|---|---|
| Nearest name in the lexicon | 145,456 | 4.2 | −0.007 | — |
| Nearest name not in the lexicon | 1,362,030 | 5.7 | −0.003 | — |
| Nearest name water-related | 89,404 | 4.4 | −0.004 | — |
| Nearest name not water-related | 1,418,082 | 5.6 | −0.003 | — |
| Slope at or below the median | 753,457 | 10.3 | −0.004 | — |
| Slope above the median | 754,029 | 0.8 | −0.001 | — |
| Built-up cover above 10 % | 100,180 | 1.3 | +0.002 | [−0.002, +0.007] |
| Built-up cover at most 10 % | 1,407,306 | 5.8 | −0.004 | [−0.011, +0.003] |
| Africa | 144,265 | 7.5 | −0.002 | [−0.006, +0.003] |
| Asia | 196,090 | 15.7 | −0.010 | [−0.028, +0.003] |
| Europe | 824,854 | 2.3 | −0.001 | [−0.015, +0.012] |
| North America | 85,031 | 4.5 | −0.001 | [−0.008, +0.005] |
| Oceania | 150,806 | 8.8 | +0.006 | [−0.009, +0.027] |
| South America | 106,440 | 5.6 | −0.005 | [−0.010, +0.0005] |
| Language: en (en, fr, sw) | 403,172 | 6.3 | +0.002 | [−0.006, +0.011] |
| Language: es (es, nah, qu) | 169,326 | 4.4 | +0.0001 | [−0.007, +0.008] |
| Language: fr (fr, mg, ng) | 122,776 | 2.8 | −0.007 | [−0.017, −0.001] |
| Language: it (de, it) | 86,262 | 1.1 | — | — |
| Language: de (de) | 83,889 | 4.3 | −0.005 | [−0.054, +0.038] |
| Language: pt (bantu, pt) | 65,887 | 4.1 | −0.016 | [−0.036, −0.002] |
| Language: ur (fa, hi) | 61,219 | 31.5 | −0.051 | [−0.150, +0.018] |
| Language: ro (ro) | 55,634 | 1.0 | — | — |
| Language: vi (vi) | 48,556 | 5.1 | −0.003 | [−0.017, +0.009] |
| Language: sv (sv) | 43,174 | 0.9 | — | — |
| Language: el (el) | 35,780 | 5.1 | −0.006 | [−0.016, +0.005] |
| Language: no (no) | 33,813 | 1.5 | +0.007 | [−0.044, +0.068] |
| Language: sq (sq) | 28,390 | 2.6 | +0.005 | [−0.095, +0.120] |
| Language: sr (sh) | 28,164 | 2.2 | +0.223 | [−0.060, +0.898] |
| Language: fa (fa) | 23,990 | 7.8 | +0.0006 | [+0.0000, +0.0012] |
| Language: lv (lv) | 20,000 | 2.1 | +0.004 | [−0.003, +0.010] |

Table A5: Secondary metrics on the primary split (pooled over held-out folds).

| Model | Brier | Log loss | PR-AUC | Top-10 % capture (%) |
|---|---|---|---|---|
| M1 (baseline model) | 0.04374 | 0.15218 | 0.327 | 51.0 |
| M3 (refitted with names) | 0.04346 | 0.15161 | 0.332 | 51.3 |
| M1s (recalibrated M1) | 0.04337 | 0.15159 | 0.325 | 50.9 |
| M3s (+ names) | 0.04314 | 0.15105 | 0.334 | 51.7 |
| M1c (+ regional composition) | 0.04309 | 0.15104 | 0.348 | 52.9 |
| M3n (+ names within areas) | 0.04312 | 0.15113 | 0.348 | 52.9 |

Table A6: The confirmatory analysis with the frozen code (lexicon not applied to Norwegian names) and with the corrected reading (amendment 4): $R$ of the within-area comparison, 95 % interval and verdict.

| Split | $R$ (frozen) | 95 % | Verdict | $R$ (corrected) | 95 % | Verdict |
|---|---|---|---|---|---|---|
| location groups held out (primary) | −0.003 | [−0.013, +0.004] | equivalent-null | −0.004 | [−0.015, +0.003] | equivalent-null |
| countries held out | +0.001 | [−0.005, +0.011] | equivalent-null | +0.0005 | [−0.003, +0.005] | equivalent-null |
| time held out | −0.001 | [−0.027, +0.023] | undefined-denominator | −0.013 | [−0.039, +0.008] | undefined-denominator |
| after the language model's cut-off | −0.002 | [−0.007, +0.002] | equivalent-null | −0.003 | [−0.011, +0.003] | equivalent-null |

Table A7: Second-stage coefficients on the standardised scale (post hoc, amendment 5): mean over the ten folds of the primary split, with the smallest and largest fold in parentheses. M3n: the name features; M3np: the same features of names reassigned among settlements; M3d: the features centred within each AOI; M1c: the regional composition (AOI means). The reading of Norwegian names is the corrected one (amendment 4). The indicator that the lexicon covers a language of the country is one for every analysed cell with the real names, so its standardised coefficient is zero and the second stage has effectively fourteen name features.

| Feature | M3n | M3np (placebo) | M3d | M1c (composition) |
|---|---|---|---|---|
| logit of the terrain prediction | +1.925 (+1.769, +2.088) | +1.924 (+1.778, +2.085) | +1.969 (+1.883, +2.092) | +1.924 (+1.769, +2.087) |
| water, nearest name | +0.006 (−0.009, +0.015) | +0.051 (−0.001, +0.108) | +0.003 (−0.003, +0.011) | −0.015 (−0.242, +0.076) |
| wetland, nearest name | +0.001 (−0.007, +0.006) | −0.023 (−0.030, −0.017) | +0.001 (−0.004, +0.005) | −0.058 (−0.164, +0.083) |
| riverine, nearest name | −0.003 (−0.012, +0.001) | +0.027 (−0.004, +0.040) | −0.002 (−0.010, +0.005) | +0.145 (+0.003, +0.231) |
| crossing, nearest name | +0.001 (−0.013, +0.009) | −0.018 (−0.031, −0.004) | +0.005 (−0.007, +0.010) | +0.014 (−0.130, +0.156) |
| lowland, nearest name | −0.004 (−0.022, +0.011) | −0.025 (−0.042, −0.010) | −0.003 (−0.013, +0.008) | −0.260 (−0.365, −0.170) |
| highland, nearest name | +0.010 (−0.012, +0.027) | −0.033 (−0.051, −0.005) | +0.010 (−0.004, +0.027) | +0.131 (+0.017, +0.382) |
| water, three nearest (weighted) | −0.036 (−0.061, −0.019) | −0.059 (−0.089, −0.038) | −0.014 (−0.027, −0.000) | −0.271 (−0.453, +0.098) |
| wetland, three nearest (weighted) | −0.004 (−0.011, +0.004) | −0.020 (−0.043, −0.006) | −0.007 (−0.013, +0.001) | −0.073 (−0.234, +0.083) |
| riverine, three nearest (weighted) | +0.023 (+0.003, +0.035) | −0.017 (−0.028, −0.004) | +0.025 (+0.001, +0.038) | −0.037 (−0.174, +0.222) |
| crossing, three nearest (weighted) | −0.029 (−0.041, −0.016) | +0.003 (−0.008, +0.012) | −0.028 (−0.040, −0.015) | −0.033 (−0.138, +0.162) |
| lowland, three nearest (weighted) | −0.045 (−0.060, −0.025) | −0.045 (−0.082, −0.015) | −0.032 (−0.041, −0.023) | +0.213 (+0.090, +0.324) |
| highland, three nearest (weighted) | +0.019 (−0.012, +0.031) | −0.047 (−0.073, −0.012) | +0.023 (−0.004, +0.038) | +0.136 (−0.143, +0.234) |
| water-related, nearest name | +0.013 (+0.006, +0.023) | +0.037 (+0.015, +0.057) | +0.007 (+0.002, +0.013) | −0.123 (−0.195, −0.067) |
| lexicon covers the language | +0.000 (+0.000, +0.000) | +0.007 (+0.000, +0.011) | +0.000 (+0.000, +0.000) | +0.000 (+0.000, +0.000) |
| character-model score | +0.010 (−0.008, +0.026) | −0.017 (−0.090, +0.020) | +0.007 (−0.006, +0.019) | +0.224 (+0.175, +0.292) |

Table A8: Interval coverage in the pre-specified null simulations (amendment 2), the $p$ values required by the protocol (amendment 6, addendum 1), the spatial correlation of held-out residuals and reference points for the baseline model (primary split). The CSI maps use the observed flooded count of each AOI and are therefore optimistic. TOST $p$: no bootstrap replicate fell outside the margin, so $p$ is below $1/B$ for $B$ replicates (the value in brackets).

| Quantity | Value |
|---|---|
| Null simulations: coverage of the 90 % / 95 % interval (mean $R$) | 92 % / 97 % |
| Null simulations: coverage of zero, 90 % / 95 % | 84 % / 92 % |
| Null simulations: between-replicate SD of $R$ (robust SD; outlying replicates); median bootstrap SE of $R$ | 0.0045 (0.0021; 13); 0.0021 |
| Null simulations: verdicts | equivalent-null 91, inconclusive 1, negative 7, not-meaning 1 |
| H2: TOST $p$ | $< 1/B$ (0.00025) |
| H3: one-sided $p$ ($\Delta_0 > \Delta_4$); Spearman $\rho$ and exact one-sided $p$ | 0.86; -0.2, 0.39 |
| H4: TOST $p$; one-sided $p$ for an improvement | $< 1/B$ (0.00025); 0.66 |
| Holm-adjusted $p$, H3 / H4 | 1.00 / 1.00 |
| Correlation of held-out M1 residuals within AOIs by distance (km) | 0-0.5: 0.50; 0.5-1: 0.29; 1-2: 0.15; 2-5: 0.04; 5-10: 0.00; 10-20: -0.02 |
| CSI pooled over all cells of maps flagging the observed flooded share per AOI: mean HAND (Table S10) / M1 / M3n | 0.25 / 0.30 / 0.30 |
| Brier score: constant (training rate) / M0 / M1 | 0.0523 / 0.0513 / 0.0437 |

# Exploratory checks of the verdict

Table A9: Corrected equivalence test on the ratio (exploratory, amendment 13, item B1). $L_{\pm} = \Delta \mp 0.05D$ with $\Delta$ the change in Brier score from names and $D$ the baseline gain, computed on the same resamples; equivalence when the one-sided 95 % upper limit of $L_+$ lies below zero and the lower limit of $L_-$ above zero. Fieller: the 90 % interval of $R$ from the variances and covariance of the resamples (normal approximation). (a) 4,000 resamples of location groups with the fitted models held fixed; (b) 500 resamples of location groups with the second stage refitted (corrected code); (c) 500 two-stage resamples, countries then activations within each drawn country, with the second stage refitted; (d) (c) plus the variance of $L_{\pm}$ over 13 assignments of location groups to folds, corrected for overlapping training sets (Nadeau–Bengio; normal approximation). The lower block repeats the double resampling of the protocol rule's operating characteristics for the corrected test (a).

| Resampling | $R$ | $L_+$ upper | $L_-$ lower | Fieller 90 % | Equivalent |
|---|---|---|---|---|---|
| (a) Confirmatory run, models fixed | −0.003 | $-2.4 \times 10^{-4}$ | $1.8 \times 10^{-4}$ | −0.013 to 0.003 | yes |
| (a) Corrected code, models fixed | −0.004 | $-2.5 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | −0.015 to 0.004 | yes |
| (a) H4, nested | −0.003 | $-1.7 \times 10^{-4}$ | $9.8 \times 10^{-5}$ | −0.028 to 0.010 | yes |
| (a) H4, total | 0.027 | $2.5 \times 10^{-4}$ | $3.8 \times 10^{-4}$ | 0.003 to 0.161 | no |
| (a) Countries held out | 0.001 | $-6.5 \times 10^{-5}$ | $9.2 \times 10^{-5}$ | −0.008 to 0.022 | yes |
| (b) Location groups, second stage refitted | −0.004 | $-2.1 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | −0.025 to 0.016 | yes |
| (c) Activations within countries, refitted | −0.004 | $-1.9 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | −0.042 to 0.031 | yes |
| (d) As (c), with fold-assignment variance | — | $-9.6 \times 10^{-5}$ | $3.9 \times 10^{-5}$ | −0.042 to 0.031 | yes |
| *Share of samples declared equivalent with the true $R$ set to −0.05 / 0 / +0.05 (models fixed; Sect. 3.2 of the paper)* | | | | | |
| protocol's rule, uniform shift | 32 % / 99.9 % / 31 % | | | | |
| corrected test (a), uniform shift | 3.1 % / 98.6 % / 2.6 % | | | | |
| protocol's rule, shift proportional to the baseline gain | 0.0 % / 99.9 % / 0.0 % | | | | |
| corrected test (a), shift proportional to the baseline gain | 15 % / 98.6 % / 2.7 % | | | | |
| *Other measures: the change in Brier units and the increment in Brier skill over a constant forecast ($R$ over a constant forecast, with one definition, in Table A10)* | | | | | |
| Primary, confirmatory run | — | ΔBS $-2.6 \times 10^{-5}$ | | skill $-5.1 \times 10^{-4}$ | |
| Primary, corrected code | — | ΔBS $-3.1 \times 10^{-5}$ | | skill $-5.8 \times 10^{-4}$ | |
| H4, nested | — | ΔBS $-3.8 \times 10^{-5}$ | | skill $-7.3 \times 10^{-4}$ | |
| H4, total | — | ΔBS $3.0 \times 10^{-4}$ | | skill $5.7 \times 10^{-3}$ | |

Table A10: The equivalence test on the ratio (exploratory, amendment 13, item B1, reported again under amendment 14, item B1). $L_{\pm} = \Delta \mp 0.05D$, with $\Delta$ the change in Brier score from names and $D$ the baseline gain, computed on the same resamples. Each limit is given twice: the 95 % (upper) or 5 % (lower) percentile of the resamples, as the corrected test was planned, and the normal approximation $\hat{L} \pm 1.645\,\mathrm{sd}$ from the variance of the resamples; the two differ where the resampling distribution has heavy tails. TOST $p$: the larger of the two one-sided normal-approximation $p$ values. Fieller: the 90 % interval of $R$ from the variances and covariance of the resamples; its upper end is the one-sided 95 % upper limit of $R$. Injected odds ratio: the effect on all ground nearest a water-related name whose mean gain reaches the upper limit, range over the four no-effect references (estimator scale in brackets). (a) 4,000 resamples; (b), (c) and the pigeonhole row 500 resamples with the second stage refitted; (d) adds the Nadeau–Bengio variance over 13 fold assignments to (c), so only the normal approximation is defined. Amendment 11 saved only the percentile interval of the ratio. Monte Carlo standard deviation of the percentile upper limit of $L_+$ over 1,000 resamples of the replicates: (a) $3.1 \times 10^{-6}$, (b) $1.1 \times 10^{-5}$, (c) $1.6 \times 10^{-5}$. Climatology (training-side constant) Brier score 0.0523: the margin is 0.72 % of it and the upper limit of (d) 0.45 %.

| Resampling | $L_+$ upper: pct / normal | $L_-$ lower: pct / normal | TOST $p$ | Fieller 90 % | Upper $R$ | Injected OR |
|---|---|---|---|---|---|---|
| Amendment 11: activations × countries, refitted (percentile of the ratio) | — | — | — | −0.055 to 0.036 | 0.036 | 2.06–2.25 |
| (a) Location groups, models fixed | $-2.5 \times 10^{-4}$ / $-2.3 \times 10^{-4}$ | $1.7 \times 10^{-4}$ / $1.6 \times 10^{-4}$ | 0.001 | −0.015 to 0.004 | 0.004 | 1.21–1.51 (1.31–1.75) |
| (b) Location groups, second stage refitted | $-2.1 \times 10^{-4}$ / $-1.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ / $1.4 \times 10^{-4}$ | 0.003 | −0.025 to 0.016 | 0.016 | 1.58–1.82 (1.85–2.20) |
| (c) Activations within countries, refitted | $-1.9 \times 10^{-4}$ / $-9.6 \times 10^{-5}$ | $1.1 \times 10^{-4}$ / $4.0 \times 10^{-5}$ | 0.03 | −0.042 to 0.031 | 0.031 | 1.93–2.16 (2.37–2.70) |
| (d) As (c), with fold-assignment variance | — / $-9.6 \times 10^{-5}$ | — / $3.9 \times 10^{-5}$ | 0.03 | −0.042 to 0.031 | 0.031 | 1.93–2.16 (2.37–2.70) |
| Pigeonhole: location groups × activations, refitted | $-5.4 \times 10^{-5}$ / $7.2 \times 10^{-5}$ | $2.1 \times 10^{-5}$ / $-1.7 \times 10^{-4}$ | 0.13 | −0.096 to 0.067 | 0.067 | 2.67–2.83 (3.45–3.68) |
| *$R$ with the baseline gain measured over a constant forecast (Y3: the Brier score of the constant minus that of the recalibrated baseline model M1s; the only definition used in this Supplement) and the Brier skill increment over the constant* | | | | | | |
| Primary split, confirmatory run | $R = -0.003$ (−0.009 to 0.002) | | | skill $-5.1 \times 10^{-4}$ | | |
| H4, nested (constant: flooded share of the test cells) | $R = -0.012$ (−0.057 to 0.020) | | | skill $-7.3 \times 10^{-4}$ | | |
| H4, total (constant: flooded share of the test cells) | $R = 0.098$ (0.016 to 0.187) | | | skill $5.7 \times 10^{-3}$ | | |

Table A11: Robustness of the corrected test to heavy-tailed cluster contributions (exploratory, amendment 14, items B2 and B3), corrected code, primary split, fitted models held fixed. $L_+$ upper and $L_-$ lower: one-sided 95 % limits (normal approximation from the jackknife variance; bootstrap-$t$ for the studentized row). Fieller: 90 % interval of $R$ (from the jackknife variance, or from 4,000 location-group resamples for the weighted rows). Operating characteristics: 1,000 outer and 500 inner two-stage resamples (countries, then activations within each drawn country); Monte Carlo standard errors in brackets. The last block repeats the double resampling of location groups (2,000 outer and 1,000 inner resamples). An earlier run with other random draws (amendment 12, item B1) gave the protocol's rule 32 % and 33 % at the two margins. Pigeonhole bootstrap (Table A10): 28 of its 500 refitted resamples have a robust $z$ of the change in Brier score above 4, and without them its standard deviation falls from $2.5 \times 10^{-4}$ to $1.2 \times 10^{-4}$; 77 % of the location groups hold a single activation and 70 % of the activations lie within one location group (described after the result, amendment 14, addendum 4).

| Variant | $R$ | $L_+$ upper | $L_-$ lower | Fieller 90 % |
|---|---|---|---|---|
| Jackknife over location groups (195) | $-0.004$ | $-2.3 \times 10^{-4}$ | $1.6 \times 10^{-4}$ | $-0.015$ to 0.004 |
| Jackknife over activations (188) | $-0.004$ | $-2.2 \times 10^{-4}$ | $1.6 \times 10^{-4}$ | $-0.015$ to 0.004 |
| Studentized bootstrap over location groups (4,000) | — | $-2.7 \times 10^{-4}$ | $1.9 \times 10^{-4}$ | — |
| Without the two most influential groups (PK, IR) | 0.0001 | $-2.1 \times 10^{-4}$ | $2.2 \times 10^{-4}$ | $-0.004$ to 0.005 |
| Equal weight per location group | $-0.003$ | $-2.9 \times 10^{-4}$ | $2.3 \times 10^{-4}$ | $-0.011$ to 0.003 |
| Equal weight per AOI | $-0.001$ | $-2.9 \times 10^{-4}$ | $2.8 \times 10^{-4}$ | $-0.005$ to 0.003 |
| Equal weight per activation | 0.001 | $-2.4 \times 10^{-4}$ | $2.5 \times 10^{-4}$ | $-0.003$ to 0.005 |
| *Effective number of clusters $G^*$ (Carter et al., 2017) for the location-group contributions* | | | | |
| Primary split, corrected code | $G = 195$; $G^* = 1.7$ for the change from names, 2.7 for the baseline gain; five groups hold 88 % of the variance | | | |
| Primary split, confirmatory run | $G = 195$; $G^* = 2.3$ for the change from names, 2.7 for the baseline gain; five groups hold 85 % of the variance | | | |
| Activations kept whole | $G = 106$; $G^* = 2.2$ for the change from names, 3.4 for the baseline gain; five groups hold 96 % of the variance | | | |
| Countries held out | $G = 53$; $G^* = 1.4$ for the change from names, 2.7 for the baseline gain; five groups hold 92 % of the variance | | | |
| *Share of samples declared equivalent (L rule) or non-superior (upper limit only) with the true $R$ set to $-0.05$ / 0 / $+0.05$; activations within countries, models fixed* | | | | |
| L rule, uniform shift | 1.7 % (0.4) / 91.6 % (0.9) / 1.7 % (0.4) | | | |
| upper limit only, uniform shift | 100.0 % (0.0) / 99.8 % (0.1) / 1.7 % (0.4) | | | |
| L rule, shift proportional to the baseline gain | 17.7 % (1.2) / 91.6 % (0.9) / 0.7 % (0.3) | | | |
| upper limit only, shift proportional to the baseline gain | 100.0 % (0.0) / 99.8 % (0.1) / 0.7 % (0.3) | | | |
| *The same with location groups resampled (amendment 13, confirmatory run): share declared equivalent at $R = -0.05$ / 0 / $+0.05$* | | | | |
| protocol's rule, uniform shift | 32.4 % / 99.9 % / 31.2 % | | | |
| L rule, uniform shift | 3.1 % / 98.6 % / 2.6 % | | | |
| protocol's rule, proportional shift | 0.0 % / 99.9 % / 0.0 % | | | |
| L rule, proportional shift | 14.9 % / 98.6 % / 2.7 % | | | |

Table A12: Decomposition of the names increment over the recalibrated baseline model (exploratory, amendment 14, item B5). $\mathrm{BS(M1s)} - \mathrm{BS(M3n)} = [\mathrm{BS(M1s)} - \mathrm{BS(M1c)}] + [\mathrm{BS(M1c)} - \mathrm{BS(M3n)}]$, each divided by the baseline gain of the split: the regional composition (M1s to M1c), the names given the composition (M1c to M3n, the primary estimand), the names centred within AOIs, which contain no composition (M1s to M3d), and the total (M1s to M3s). Brackets: 90 % Fieller intervals of the corrected test (a), 4,000 location-group resamples (unbounded where the baseline gain is not distinguishable from zero). Confirmatory rows take M3d from the run with centred features on the same cells.

| Split | BS(M1s) | BS(M1c) | BS(M3n) | Composition | Names | Centred (M3d) | Total (M3s) |
|---|---|---|---|---|---|---|---|
| Primary (corrected) | 0.04337 | 0.04305 | 0.04308 | 0.043 | $-0.004$ | $-0.004$ ($-0.010$ to $-0.001$) | 0.031 ($-0.060$ to 0.143) |
| Language-model cut-off (corrected) | 0.04941 | 0.04988 | 0.04991 | $-0.042$ | $-0.003$ | $-0.0002$ ($-0.003$ to 0.007) | 0.003 ($-0.017$ to 0.079) |
| Primary (confirmatory) | 0.04337 | 0.04309 | 0.04312 | 0.037 | $-0.003$ | $-0.004$ ($-0.010$ to $-0.001$) | 0.031 ($-0.058$ to 0.142) |
| Activations kept whole (confirmatory) | 0.04767 | 0.05176 | 0.05159 | $-0.430$ | 0.018 | $-0.003$ ($-0.005$ to $-0.001$) | $-0.032$ ($-0.135$ to 0.091) |
| Countries (confirmatory) | 0.04789 | 0.04849 | 0.04848 | $-0.085$ | 0.001 | $-0.003$ ($-0.020$ to $-0.001$) | $-0.037$ ($-0.257$ to 0.099) |
| Time (confirmatory) | 0.04605 | 0.04677 | 0.04677 | $-0.311$ | $-0.001$ | $-0.004$ (unbounded) | 0.007 (unbounded) |
| Language-model cut-off (confirmatory) | 0.04941 | 0.04974 | 0.04976 | $-0.030$ | $-0.002$ | $-0.0003$ ($-0.004$ to 0.007) | 0.001 ($-0.018$ to 0.073) |

Table A13: Cluster runs of amendment 14 (exploratory, items C1 and C3). Observable ground: cells whose 10 m built-up and tree cover sum to 0.5 or more left out of evaluation, the cap drawn from the rest. Language-model reading: the total increment on floods after the model's training cut-off, with and without country indicators in the reference model. $R$ with the 90 % interval of the protocol's rule and its verdict; last column: 90 % Fieller interval of the corrected test (a).

| Comparison | Cells | Baseline gain | $R$ (90 %) | Verdict | Corrected (a) |
|---|---|---|---|---|---|
| Observable ground only: M1c to M3n | 1,303,026 | $9.58 \times 10^{-3}$ | 0.005 ($-0.001$ to 0.011) | equivalent-null | $-0.002$ to 0.016 |
| Language-model reading: M1s to M3rs | 268,762 | $1.12 \times 10^{-2}$ | 0.027 (0.004 to 0.051) | positive-small | 0.003 to 0.147 |
| Language-model reading with country indicators: M1k to M3rk | 268,762 | $1.12 \times 10^{-2}$ | 0.019 ($-0.005$ to 0.053) | inconclusive | $-0.044$ to 0.038 |
| Brier scores on the cut-off split | M1s 0.0494, M3rs 0.0491, M1k 0.0543, M3rk 0.0541 | | | | |

Table A14: The names increment by AOI (exploratory, amendment 14, item B15), corrected code, primary split: change in Brier score from M1c to M3n (positive when names improve) and change in the area under the ROC curve (AOIs with both classes), by continent and flood type; and $R$ with each of the five activations of largest jackknife influence left out (fitted models held fixed). Continents follow the country of the settlements nearest most of an AOI's cells (GeoNames); the paper and Table S1 follow the continent of the EMS activation, and the two differ for two AOIs between Europe and Asia.

| AOIs | $n$ | Median ΔBS | Share > 0 | Median ΔAUC | Share > 0 |
|---|---|---|---|---|---|
| All AOIs | 500 | $-2.5 \times 10^{-6}$ | 47 % | 0.0000 | 49 % |
| AF | 40 | $-1.5 \times 10^{-5}$ | 40 % | 0.0000 | 53 % |
| AS | 55 | $4.8 \times 10^{-5}$ | 58 % | 0.0000 | 51 % |
| EU | 328 | $-1.8 \times 10^{-6}$ | 48 % | 0.0000 | 49 % |
| NA | 16 | $-1.2 \times 10^{-5}$ | 25 % | −0.0005 | 31 % |
| OC | 45 | $1.9 \times 10^{-6}$ | 51 % | 0.0000 | 49 % |
| SA | 16 | $-1.9 \times 10^{-5}$ | 31 % | −0.0001 | 44 % |
| Typed | 346 | $-1.7 \times 10^{-6}$ | 49 % | 0.0000 | 48 % |
| Untyped (inland, counted as riverine) | 154 | $-3.8 \times 10^{-6}$ | 44 % | 0.0000 | 50 % |
| $R$ without each of the five most influential activations: EMSR838 0.0002, EMSR097 −0.003, EMSR419 −0.005, EMSR629 −0.005, EMSR567 −0.005 | | | | | |

Table A15: Direction of the exploratory interval estimates that exclude the no-effect value, in the final census (exploratory, amendment 14, item B10). Favouring names: more flooding (or lower, wetter sites) near water-related names, less near highland names, or a positive $R$. Rows that the rule could not classify (mostly estimates without a class in their key) and rows without a direction are counted together.

| Family | Exclude null | Favour names | Against names | Unclassified |
|---|---|---|---|---|
| Flooding (labels or Brier score) | 79 | 10 | 57 | 12 |
| Settlement sites (no flood labels) | 98 | 66 | 2 | 30 |
| Other terms (bands, base logit) | 125 | 0 | 0 | 125 |

Table A16: Models of the post hoc analyses. All use the second stage of the primary comparison (Table 1 of the paper); the amendment is the one that planned the model.

| Code | Information | Amendment |
|---|---|---|
| M3d | logit(M1) + name features centred within each AOI, without composition terms | 2 |
| M1co, M3o | as M1c and M3n with the HAND of the nearest settlement's site in place of the names | 5 |
| M1cL, M3oL | as M1c and M3n with the flooded share within 500 m of the nearest settlement in the same event (the cell itself left out) | 6 |
| M1cG, M3oG | as M1c and M3n with recurrent surface water within 500 m of the nearest settlement (occurrence and seasonality, 1984–2021) | 6 |
| M1cS, M3nS, M3nSp | as M1c, M3n and M3np with the strict reading (no springs or wells) | 6 |
| T0$^{-}$ | T0 without river distances and without the water and wetland classes of land cover (rung of the ladder) | 6 |
| M1cEc, M3oEc | as M1c and M3n with the share of other EMS events (more than 30 days apart) in which the cell itself flooded, and whether there is such an event | 7 |
| M1cE, M3oE | the same share averaged over the cells within 500 m of each settlement and assigned through the nearest settlement | 7 |
| M1cF, M3nF, M3nFp | as M1c, M3n and M3np with the floodplain reading (riverine and wetland classes) | 7 |
| M1 + c(q), M1 + c(q) + q | reference and names models of a reassignment $q$, with the regional composition recomputed from the reassigned names | 7 |
| M1cW, M3oW | as M1c and M3n with recurrent surface water of the cell itself (occurrence, seasonality and maximum extent, 1984–2021, averaged over the cell) | 8 |

Table A17: Post hoc checks of the within-area comparison (M1c to M3n or its analogue): $R$ with its 90 % cluster-bootstrap interval and the verdict of the protocol's decision table. Amendment: the amendment that planned the check; code: the frozen code of the confirmatory run or the code with the corrected reading of Norwegian names. The measure of the baseline gain is that of each run (for the European AOIs, that of these AOIs).

| Check | Amendment | Code | $R$ | 90 % interval | Verdict |
|---|---|---|---|---|---|
| Flood events held out whole | 2 | frozen | +0.018 | [+0.001, +0.039] | not-meaning |
| Flood events held out whole | 7 | corrected | +0.004 | [−0.004, +0.014] | equivalent-null |
| Flood traces counted as not flooded | 5 | corrected | −0.001 | [−0.007, +0.004] | equivalent-null |
| Flooded share over land only | 5 | corrected | +0.003 | [−0.002, +0.009] | equivalent-null |
| Partly flooded cells excluded | 7 | corrected | −0.004 | [−0.010, +0.001] | equivalent-null |
| Every AOI weighted equally | 6 | frozen | −0.0003 | [−0.004, +0.004] | — |
| European AOIs alone, own baseline gain | 7 | corrected | −0.0005 | [−0.013, +0.013] | equivalent-null |
| Without the Americas and Oceania | 5 | corrected | −0.006 | [−0.015, +0.002] | — |
| Strict reading | 6 | corrected | −0.003 | [−0.006, +0.0004] | equivalent-null |
| Gazetteer cleaned of city sections and duplicates | 6 | corrected | −0.003 | [−0.007, +0.0004] | equivalent-null |
| Without AOIs lacking a hydrography layer | 6 | corrected | −0.002 | [−0.006, +0.004] | equivalent-null |
| Without AOIs lacking a flood type | 6 | corrected | −0.009 | [−0.019, −0.001] | negative |
| Language-model reading, primary split | 6 | corrected | −0.001 | [−0.008, +0.005] | equivalent-null |
| Floodplain reading (riverine and wetland classes) | 7 | corrected | −0.002 | [−0.004, +0.0003] | equivalent-null |
| Cleaned lexicon (two-letter affixes and English suffixes removed) | 8 | corrected | −0.004 | [−0.009, +0.001] | equivalent-null |
| GFM labels, existing predictions rescored | 3 | frozen | −0.003 | [−0.020, +0.011] | equivalent-null |

Table A18: The primary comparison in subsets of cells and under variants of the fold assignment (post hoc). For subsets, $R$ is the names increment in the subset divided by the baseline gain in the same subset (as for the H5 strata); the 90 % interval comes from the cluster bootstrap over location groups restricted to the subset, and the last column gives the equivalence test alone. For variants, the last column gives the label of the pre-specified decision table and the equivalence test; the run with one seed instead of five is the pre-specified sensitivity analysis 5 (protocol §7-5). Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 %. Distance bands up to 500 m, 0.5–1 km and beyond 1 km, the classes, the sensors, the fold reassignments and the activation grouping were planned in amendment 2 before they were run. The baseline gain within a class differs between classes by up to about seventeen-fold, so the verdicts by class depend strongly on their own denominators; Table A42 compares classes on the Brier scale. [a]Not planned (added while the analysis was written, or a subset of such an analysis). Values of $R$ for subsets with a small baseline gain (for example single classes of names) are unstable; the increments by class on the Brier scale are in the table of increments by class.

| Subset or variant | Cells | Units | Baseline ΔBS | $R$ | 90 % interval | Verdict |
|---|---|---|---|---|---|---|
| *Distance from the cell to its nearest settlement* | | | | | | |
| up to 500 m | 160,239 | 195 | $6.48 \times 10^{-3}$ | −0.003 | [−0.011, +0.004] | equivalent |
| up to 250 m[a] | 44,030 | 195 | $6.65 \times 10^{-3}$ | −0.001 | [−0.008, +0.005] | equivalent |
| 250–500 m[a] | 116,209 | 195 | $6.41 \times 10^{-3}$ | −0.004 | [−0.012, +0.004] | equivalent |
| 0.5–1 km | 312,024 | 195 | $6.67 \times 10^{-3}$ | −0.002 | [−0.011, +0.006] | equivalent |
| more than 1 km | 1,035,223 | 195 | $7.99 \times 10^{-3}$ | −0.004 | [−0.011, +0.002] | equivalent |
| up to 500 m, nearest name water-related[a] | 10,082 | 134 | $5.84 \times 10^{-3}$ | +0.060 | [−0.036, +0.162] | not equivalent |
| up to 500 m, built-up and tree cover removed[a] | 111,426 | 193 | $8.33 \times 10^{-3}$ | −0.005 | [−0.012, +0.003] | equivalent |
| up to 500 m, water-related, built-up and tree cover removed[a] | 6,945 | 130 | $7.70 \times 10^{-3}$ | +0.051 | [−0.050, +0.155] | not equivalent |
| *Class of the nearest name* | | | | | | |
| water-related (any) | 89,404 | 146 | $1.25 \times 10^{-2}$ | −0.004 | [−0.023, +0.016] | equivalent |
| water | 35,814 | 118 | $2.65 \times 10^{-2}$ | +0.011 | [−0.001, +0.025] | equivalent |
| wetland | 2,857 | 51 | $1.54 \times 10^{-3}$ | −0.499 | [−1.262, −0.0002] | not equivalent |
| riverine | 13,841 | 85 | $5.81 \times 10^{-3}$ | −0.244 | [−0.509, +0.004] | not equivalent |
| crossing | 13,918 | 83 | $1.97 \times 10^{-3}$ | +0.106 | [−0.135, +0.395] | not equivalent |
| lowland | 24,552 | 101 | $2.71 \times 10^{-3}$ | +0.074 | [−0.090, +0.265] | not equivalent |
| highland | 58,021 | 130 | $8.62 \times 10^{-3}$ | −0.013 | [−0.038, +0.013] | equivalent |
| *Post-event imagery of the delineation* | | | | | | |
| radar only | 839,258 | 122 | $5.41 \times 10^{-3}$ | −0.006 | [−0.024, +0.008] | equivalent |
| optical only | 174,932 | 37 | $2.19 \times 10^{-2}$ | +0.0003 | [−0.003, +0.004] | equivalent |
| not recorded (mostly schema generations 0–1) | 493,296 | 65 | $6.10 \times 10^{-3}$ | −0.004 | [−0.012, +0.004] | equivalent |
| *Fold assignment, grouping and seeds (frozen code; verdict of the decision table)* | | | | | | |
| pre-specified folds | 1,507,486 | 195 | $7.55 \times 10^{-3}$ | −0.003 | [−0.011, +0.003] | equivalent-null (TOST holds) |
| folds reassigned (seed 1) | 1,507,486 | 195 | $7.50 \times 10^{-3}$ | −0.005 | [−0.012, +0.0003] | equivalent-null (TOST holds) |
| folds reassigned (seed 2) | 1,507,486 | 195 | $7.04 \times 10^{-3}$ | +0.005 | [−0.001, +0.013] | equivalent-null (TOST holds) |
| AOIs of the same activation grouped | 1,507,486 | 106 | $9.53 \times 10^{-3}$ | +0.018 | [+0.001, +0.039] | not-meaning (TOST holds) |
| one seed instead of five | 1,507,486 | 195 | $5.69 \times 10^{-3}$ | −0.004 | [−0.011, +0.002] | equivalent-null (TOST holds) |

Table A19: The verdict in more detail (exploratory, amendment 8). Achieved bound: the smallest symmetric margin within which the 90 % interval of $R$ lies. Location groups needed: the multiple of the present number of groups at which the 90 % interval would lie within ±0.05 (equivalence) or the lower 95 % limit would exceed zero (positive), if the half-width shrinks with the square root of the number of groups and the point estimate stays; — where the point estimate lies outside the margins. These multiples are power calculations at the observed point estimate and add nothing to the intervals; the paper no longer uses them. Permutations: 200 reassignments of names among the settlements of each AOI with the regional composition rebuilt. Refitted interval: 500 bootstrap replicates of location groups in which the second stage of both models is refitted with the replicate's weights, against the same replicates with the fitted models held fixed. Effect at the margin: an odds ratio of 2.34 for all ground nearest a water-related name injected into the labels, every model refitted, and each replicate judged by the protocol's decision table without the placebo condition.

| Quantity | Value |
|---|---|
| Achieved equivalence bound, frozen code (verdict): $R$; Brier units | 0.011; $8.3 \times 10^{-5}$ |
| Achieved equivalence bound, corrected reading: $R$; Brier units | 0.013; $9.5 \times 10^{-5}$ |
| largest over five measures of the baseline gain and two bootstraps (frozen code) | 0.046 (two-way, Y1) |
| Location groups needed, total increment: equivalence; positive (multiple of 195) | 20.2; 11.3 |
| Location groups needed, flexible models refitted: equivalence; positive (multiple of 195) | 45.7; 7.0 |
| Location groups needed, within 500 m of water-related names (not planned): equivalence; positive (multiple of 134) | —; 3.8 |
| Permutations with rebuilt composition: one-sided $p$; mean and 95th percentile of $R$ | 0.72; −0.002, 0.004 |
| label with the permutation distribution as the placebo | unchanged (equivalent-null) |
| Corrected run, second stage refitted in each replicate: 90 % (95 %) interval of $R$ | −0.017 to 0.013 (−0.024 to 0.019) |
| Corrected run, same replicates, models fixed: 90 % (95 %) interval of $R$ | −0.013 to 0.003 (−0.014 to 0.004) |
| standard deviation of $R$, refitted over fixed | 2.31 |
| Effect at the margin injected, every model refitted: replicates judged equivalent (Wilson 95 %) | 0 of 40 (0.0 %–8.8 %) |
| labels; mean $R$ | positive-small 40; 0.040 |

Table A20: The skill of the held-out predictions of the corrected run with and without the names, decomposed (exploratory; amendment 9). Murphy decomposition (Murphy, 1973) over ten bins of predicted probability (the binned form omits the within-bin variance and covariance terms, so reliability minus resolution plus uncertainty does not reproduce the Brier score exactly); within AOIs: each model's predictions shifted on the logit scale so that its mean in each AOI equals the observed flooded fraction (479 AOIs with both flooded and dry cells), and the comparison and measure of the baseline gain recomputed; maps: in each AOI, the cells with the highest predicted probability (for HAND, the lowest HAND, ties broken by mean HAND and then at random) flagged as flooded, as many as were observed flooded (479 AOIs). Differences of the decomposition and of the critical success index: M3n minus M1c; the names raised the index in 16 % of the AOIs and lowered it in 23 %. M1c: without names; M3n: with names; M1 ranks cells within AOIs as M1c does. Intervals: 95 % from a cluster bootstrap over location groups (4,000 replicates) unless stated.

| Quantity | M1c | M3n | Difference or reference |
|---|---|---|---|
| *Decomposition of the Brier score (Murphy, 1973)* | | | |
| Brier score | 0.04305 | 0.04308 | $3.1 \times 10^{-5}$ |
| Reliability (lower is better) | $2.0 \times 10^{-4}$ | $2.1 \times 10^{-4}$ | $8.0 \times 10^{-6}$ ($-5.7 \times 10^{-6}$ to $3.6 \times 10^{-5}$) |
| Resolution (higher is better) | $8.0 \times 10^{-3}$ | $8.0 \times 10^{-3}$ | $-1.3 \times 10^{-5}$ ($-4.2 \times 10^{-5}$ to $1.3 \times 10^{-5}$) |
| Uncertainty | $5.2 \times 10^{-2}$ | | |
| *Within AOIs (each AOI's flooded fraction given)* | | | |
| Baseline gain within AOIs | $7.5 \times 10^{-3}$ | | |
| Names increment BS(M1c) − BS(M3n) | | | $-3.1 \times 10^{-5}$ ($-5.8 \times 10^{-5}$ to $-7.2 \times 10^{-6}$) |
| $R$ (90 % interval) | | | −0.004 [−0.007, −0.001] |
| *Maps flagging the observed number of flooded cells per AOI (median; quartiles)* | | | |
| Critical success index | 0.129 (0.03–0.27) | 0.132 (0.03–0.27) | mean HAND: 0.084 (0.00–0.21) |
| Hit rate | 0.229 (0.07–0.42) | 0.233 (0.06–0.43) | mean HAND: 0.155 |
| False-alarm ratio | 0.771 (0.58–0.93) | 0.767 (0.57–0.94) | mean HAND: 0.845 |
| Difference in the critical success index | | | median 0.000 (0.000 to 0.000) |

Table A21: The primary comparison under the robustness runs of amendment 9 (exploratory; corrected code; the primary row uses the frozen subsampling seed 20261025). Cells: evaluated cells; baseline gain: BS(M0) − BS(M1) in units of $10^{-3}$; odds ratio: flooding within 500 m of water-related names relative to other names, given the held-out terrain prediction of the run, with a pairs cluster bootstrap 95 % interval (2,000 replicates); not computed for the seeds. Permanent water: cells whose mean JRC GSW occurrence at 25 points is at least 50 % were removed in every AOI (32,914 cells, 2.2 %, in 271 AOIs; 14,527 in AOIs with a hydrography layer and 18,387 in AOIs without one), in addition to the water of each product's hydrography layer. Seeds: the seed that draws cells within location groups under the cap (frozen: 20261025). Bare-earth elevation: the elevation features rebuilt from FABDEM v1-2 (Hawker et al., 2022), and HAND replaced by the heights of each cell above the nearest vertex of a HydroRIVERS river draining at least 100 and 1000 $km^2$; the location model M0 is unchanged. Its baseline model flagged the observed number of flooded cells per AOI with a median critical success index of 0.12 (with GLO-30 and HAND: 0.13); with the names, the median difference was 0.000. By the rule fixed in amendment 9, these runs do not replace the confirmatory verdict, and an odds ratio whose interval excludes one would remain an uncorroborated description.

| Run | Cells | Baseline gain | $R$ | 90 % interval | Verdict | Odds ratio (95 %) |
|---|---|---|---|---|---|---|
| Primary (corrected code) | 1,507,486 | 7.55 | −0.004 | [−0.013, +0.003] | equivalent-null | 1.01 (0.77–1.31) |
| GSW permanent water removed | 1,474,572 | 8.58 | −0.004 | [−0.009, −0.0002] | equivalent-null | 1.04 (0.78–1.38) |
| Subsampling seed 1 | 1,507,486 | 7.48 | −0.002 | [−0.006, +0.003] | equivalent-null | — |
| Subsampling seed 2 | 1,507,486 | 7.46 | +0.0004 | [−0.004, +0.005] | equivalent-null | — |
| Subsampling seed 3 | 1,507,486 | 8.12 | −0.001 | [−0.004, +0.003] | equivalent-null | — |
| FABDEM terrain (bare earth) | 1,507,486 | 6.55 | −0.002 | [−0.006, +0.003] | equivalent-null | 1.02 (0.78–1.33) |

Table A22: Robustness of the primary comparison and alternative estimands (exploratory, amendment 10). $R$ with 90 % and 95 % cluster-bootstrap intervals over location groups; in subsets the baseline gain is measured on the same subset. Sensors are those of the EMS post-event images (Sentinel-1 if any image is Sentinel-1). Part C reruns the second stage on the primary split with the corrected code and applies the decision table of the protocol, without replacing the confirmatory verdict; the JRC maps are estimated from a reanalysis that includes the event years. [a]The logit of each model's held-out prediction is shifted within each AOI so that its mean equals the observed flooded fraction, and AOIs that are all flooded or all dry are left out; the baseline gain is measured in the same way. The placebo rows were added after the GFM result had been seen (addendum 2 of amendment 10), because the decision table requires them for an interval above zero; the last column gives $R$ minus the placebo's $R$ with its 95 % interval. [b]The estimator of Eq. (2) of the paper (AOI fixed effects, distance bands and the logit of the baseline model's held-out prediction) on the cells of each subset, with a 95 % pairs cluster bootstrap over location groups (2,000 replicates); not estimated when fewer than five water-related cells within 500 m flooded. [c]The pairs bootstrap separated in some replicates (standard deviation of the log odds ratio above 3), so the cluster-robust interval is given.

| Comparison | $R$ | 90 % interval | 95 % interval | Verdict or sample |
|---|---|---|---|---|
| *Primary comparison in subsets (corrected code; base models of the full run)* | | | | |
| EMS map from Sentinel-1 | −0.018 | −0.156 to 0.010 | −0.289 to 0.015 | 160 AOIs, 77 groups |
| EMS map from other radar | −0.001 | −0.008 to 0.019 | −0.011 to 0.032 | 132 AOIs, 70 groups |
| EMS map from optical images | 0.003 | −0.002 to 0.026 | −0.003 to 0.040 | 56 AOIs, 37 groups |
| Images not recorded | −0.007 | −0.016 to 0.001 | −0.018 to 0.003 | 152 AOIs, 65 groups |
| Product schema generation 0 | −0.002 | −0.054 to 0.019 | −0.107 to 0.025 | 31 AOIs, 12 groups |
| Product schema generation 1 | −0.011 | −0.020 to −0.001 | −0.023 to 0.001 | 119 AOIs, 56 groups |
| Product schema generation 2 | 0.001 | −0.003 to 0.007 | −0.006 to 0.009 | 61 AOIs, 28 groups |
| Product schema generation 3 | −0.004 | −0.025 to 0.007 | −0.037 to 0.010 | 289 AOIs, 132 groups |
| Without the African AOIs | −0.005 | −0.020 to 0.004 | −0.025 to 0.005 | 460 AOIs, 177 groups |
| *Odds ratio within 500 m of water-related names in the same subsets*[b] | | | | |
| EMS map from Sentinel-1 | 0.90 | | 0.61–1.30 | 45 flooded cells |
| EMS map from other radar | 1.44 | | 0.87–2.42 | 86 flooded cells |
| EMS map from optical images | 0.57 | | 0.44–0.91 | 43 flooded cells |
| EMS map from images not recorded | 1.02 | | 0.62–1.60 | 187 flooded cells |
| Product schema generation 0 | 0.93 | | 0.28–3.16 (cluster-robust)[c] | 25 flooded cells |
| Product schema generation 1 | 1.14 | | 0.55–1.87 | 122 flooded cells |
| Product schema generation 2 | — | | too few flooded cells | 4 |
| Product schema generation 3 | 1.02 | | 0.80–1.36 | 210 flooded cells |
| *Alternative second stages and references (amendment 10, part C)* | | | | |
| Name features × distance bands (M1cd → M3nd) | −0.008 | −0.017 to 0.001 | −0.019 to 0.003 | equivalent-null |
| Name features × continents (M1cr → M3nr) | −0.062 | −0.108 to −0.024 | −0.117 to −0.018 | negative |
| Penalty by grouped inner cross-validation (M1ct → M3nt) | −0.013 | −0.032 to 0.005 | −0.036 to 0.008 | equivalent-null |
| Country indicators in the reference (M1kc → M3kn) | −0.005 | −0.011 to 0.001 | −0.012 to 0.003 | equivalent-null |
| Baseline model with JRC flood hazard maps (M1hc → M3hn) | −0.004 | −0.007 to −0.001 | −0.007 to −0.001 | negative |
| Regional composition after the country indicators (M1k → M1kc), share of the baseline gain | −0.029 | | −0.233 to 0.170 | |
| *Scored within AOIs, each AOI's flooded fraction given*[a] | | | | |
| EMS labels, corrected run (amendment 9, item B4) | −0.004 | −0.007 to −0.001 | −0.008 to −0.001 | negative |
| reassigned names (placebo); real minus placebo | −0.006 | | −0.010 to −0.001 | 0.001 (−0.004 to 0.007) |
| GFM labels, every model refitted (amendment 10, item B10) | 0.011 | 0.004 to 0.019 | 0.003 to 0.021 | positive-negligible |
| reassigned names (placebo); real minus placebo | −0.017 | | −0.033 to −0.005 | 0.028 (0.016 to 0.044) |

Table A23: Brier scores of the held-out predictions of each split (exploratory, amendment 10, item B9). Code: the frozen code of the confirmatory run or the corrected code; on the activation split the baseline gain is Y0 $9.53 \times 10^{-3}$, Y1 $3.94 \times 10^{-3}$, Y2 $5.12 \times 10^{-3}$, Y3 $4.75 \times 10^{-3}$, Y4 $3.93 \times 10^{-4}$; Table S7 defines the measures of the baseline gain and the bootstraps. On the activation split the reference model M1c is worse than the recalibrated baseline model M1s. Lower block: $R$ under each measure, with the 90 % interval of the cluster and the two-way bootstrap ([†]: the interval does not lie within ±0.05).

| Split | Code | M0 | M1 | M1s | M1c | M3n |
|---|---|---|---|---|---|---|
| Location groups (primary) | corrected | 0.0513 | 0.0437 | 0.0434 | 0.0430 | 0.0431 |
| Countries | frozen | 0.0565 | 0.0494 | 0.0479 | 0.0485 | 0.0485 |
| Time (from 2024) | frozen | 0.0495 | 0.0472 | 0.0460 | 0.0468 | 0.0468 |
| Language-model cut-off | corrected | 0.0611 | 0.0499 | 0.0494 | 0.0499 | 0.0499 |
| Activations kept whole | corrected | 0.0583 | 0.0488 | 0.0477 | 0.0519 | 0.0518 |
| Activations kept whole | frozen | 0.0583 | 0.0488 | 0.0477 | 0.0518 | 0.0516 |
| *$R$ on the activation split (corrected code) under each measure of the baseline gain: 90 % interval, cluster bootstrap and two-way bootstrap* | | | | | | |
| Y0: pre-specified | | 0.004 | −0.004 to 0.014 | −0.007 to 0.021 | | |
| Y1: lower 95 % bound | | 0.009 | −0.008 to 0.033 | −0.036 to 0.116[†] | | |
| Y2: recalibrated | | 0.007 | −0.007 to 0.025 | −0.012 to 0.039 | | |
| Y3: from a constant | | 0.008 | −0.007 to 0.027 | −0.013 to 0.042 | | |
| Y4: stacked | | 0.091 | −0.085 to 0.332[†] | −0.158 to 0.508[†] | | |

Table A24: Further checks of amendment 11 (exploratory). B1: the change of $R$ on the activation split between the frozen and the corrected code, by location group. B6: a 90 % interval from the corrected variance of Nadeau and Bengio (2003) over the 13 fold assignments (it reflects refitting, not the sampling of test cells). B7: 500 replicates that refit the second stage and recompute the baseline gain under each resampling. B4: the predictions of the corrected run, trained on EMS labels, scored on EMS labels. B5: labels with the flooded fraction measured over the observed area that is not permanent water. B10: the pilots, with synthetic labels calibrated to the real flooded fraction of each AOI, in which names and reassigned names improved the nested comparison alike. C1: on the country split, the names reassigned within each AOI 100 times with the regional composition recomputed from each reassignment (corrected code).

| Quantity | Value | Note |
|---|---|---|
| *Activation split: frozen and corrected code (item B1)* | | |
| $R$, frozen code; corrected code | 0.018; 0.004 | M0 and M1 identical in the two runs |
| Change of $R$; from location groups with Norwegian cells | −0.015; −0.00003 | 6 groups |
| unit g00098 of the activation split (PK, 38,140 cells) | −0.010 | |
| unit g00054 of the activation split (IR, 14,076 cells) | −0.002 | |
| unit g00076 of the activation split (PK, 20,000 cells) | −0.001 | |
| *Intervals for R on the primary split (items B6 and B7)* | | |
| Refitting over 13 fold assignments, Nadeau–Bengio, 90 % | −0.003 to 0.0004 | mean −0.001 |
| Second stage refitted, ratio bootstrap, activations × countries, 90 % | −0.055 to 0.036 | not within ±0.05 |
| Second stage refitted, ratio bootstrap, location groups, 90 % | −0.020 to 0.008 | within ±0.05 |
| *Country split: reassignments with the composition recomputed (item C1)* | | |
| $R$ of the real names (corrected code); mean of 100 reassignments (2.5–97.5 %) | 0.0005; −0.003 (−0.026 to 0.012) | |
| Percentile of the real names; one-sided $p$ | 69th; 0.31 | |
| *Domain of the GFM labels: EMS-trained predictions on EMS labels (item B4)* | | |
| Baseline gain and $R$, all cells of the GFM AOIs | $1.03 \times 10^{-2}$ ($1.83 \times 10^{-3}$ to $2.17 \times 10^{-2}$); −0.005 | 545,793 cells |
| Baseline gain and $R$, cells that GFM can assess | $1.57 \times 10^{-2}$ ($2.48 \times 10^{-3}$ to $3.44 \times 10^{-2}$); −0.006 | 234,703 cells |
| *Denominator of the flooded fraction (item B5)* | | |
| Cells partly unobserved or partly permanent water: all; within 500 m of water-related; of other names | 4.2 %; 5.0 %; 4.3 % | |
| Relabelled over the observed non-permanent-water area: changed to flooded; left out | 1,541; 38 | |
| $R$ rescored on these labels (90 %) | −0.004 (−0.016 to 0.003) | |
| *Pilot simulations before the freeze, no injected effect (item B10)* | | |
| Pilot 3, rep0000: nested increment $R$ of names; of reassigned names | 0.185; 0.153 | |
| Pilot 3, rep0001: nested increment $R$ of names; of reassigned names | 0.094; 0.095 | |

Table A25: Census of the exploratory interval estimates of amendments 2 to 10 (amendment 11, item B8): every estimate with a 95 % interval in the verification outputs v36 to v67, collected by key rules and counted once where the same value appears in several files. Distance-band and logit coefficients, near-minus-far contrasts, intervals that are not finite or inverted, and fits that separate are excluded, and so are the controls (records of surface water, recurrence, oracles and hazard maps). Excludes: the interval excludes the no-effect value (one for odds ratios, zero otherwise). The estimates are strongly dependent (many are variants of the same quantity). The first pass counted the excluded coefficients as name effects; the rules were refined before any count was used. The odds ratios within 500 m of settlements are those of every estimate whose key or record names a near-settlement quantity; the first version of this census flagged them by the key alone and missed 93 of them (an implementation error corrected in addendum 1 of amendment 12). The $q$ values are approximate: they are computed from $p$ values recovered from asymmetric intervals, and the estimates are not independent. The machine-readable list is v69_estimates.csv in the public repository.

| Family | Estimates | Excludes | Share |
|---|---|---|---|
| Name effects on flood labels or predictions | 641 | 79 | 12 % |
| R | 74 | 8 | 11 % |
| difference | 78 | 12 | 15 % |
| odds ratio | 489 | 59 | 12 % |
| Name effects on terrain or water descriptors (no flood labels) | 190 | 98 | 52 % |
| difference | 190 | 98 | 52 % |
| Odds ratios within 500 m of settlements | 198 | 24 | 12 % |
| Benjamini–Hochberg $q < 0.05$; $q < 0.10$ | | 0; 9 | |
| Benjamini–Yekutieli $q < 0.05$; $q < 0.10$ | | 0; 0 | |

Table A26: Further checks of amendment 12 (exploratory). B1: shares of resampled samples in which each decision rule declares equivalence when the true $R$ is set by shifting the names increment (uniformly over cells, or in proportion to each location group's baseline gain); the protocol's rule divides the interval of the change in Brier score by the point baseline gain, the ratio rule uses the interval of $R$. B10: the reference model refitted from the second-stage design of the corrected run. B17: held-out Brier scores of the position model and of the best constant forecast (the test cells' flooded share). B5: the thirteen outlying null replicates of the protocol's simulations. B4, B12, B13: inclusion, GFM images and flood typing. B18: the model of Eq. (2) of the paper with and without the share of built-up land at 10 m (ESA WorldCover 2021). B2: countries of the flooded cells within 500 m of water-related names in the sample of every evaluable cell. B11: the second stage with the penalty chosen by grouped inner cross-validation for the name features and the composition only, on a grid widened until the optimum is interior.

| Quantity | Value | Note |
|---|---|---|
| *Share of resampled samples declared equivalent (item B1)* | | |
| True $R = -0.05$, shift uniform over cells: protocol's rule; ratio rule | 32 %; 3 % | |
| True $R = 0$, shift uniform over cells: protocol's rule; ratio rule | 100 %; 98 % | |
| True $R = +0.05$, shift uniform over cells: protocol's rule; ratio rule | 33 %; 3 % | |
| True $R = -0.05$, shift proportional to the baseline gain: protocol's rule; ratio rule | 0 %; 15 % | |
| True $R = 0$, shift proportional to the baseline gain: protocol's rule; ratio rule | 100 %; 98 % | |
| True $R = +0.05$, shift proportional to the baseline gain: protocol's rule; ratio rule | 0 %; 3 % | |
| *Reference model on the primary split, corrected code (item B10)* | | |
| Brier score: M1s; M1c; M1c without the area mean of the character model's score | 0.04337; 0.04305; 0.04382 | |
| Correlation of sibling AOI residuals within 50 km (Fisher 95 %) | -0.74 (-0.92 to -0.29) | 12 pairs |
| *Position model against a constant forecast (item B17)* | | |
| Brier score of M0; of the best constant, location groups held out | 0.0513; 0.0523 | better |
| Brier score of M0; of the best constant, countries held out | 0.0565; 0.0523 | worse |
| Brier score of M0; of the best constant, floods from 2024 held out | 0.0495; 0.0473 | worse |
| Brier score of M0; of the best constant, floods after the language model's cut-off | 0.0611; 0.0525 | worse |
| Brier score of M0; of the best constant, activations kept whole | 0.0583; 0.0523 | worse |
| *Outlying null simulations of the protocol (item B5)* | | |
| Replicates led by location group g00024 (BD); its share of $R$ | 5 | 0.84 to 1.10 |
| Replicates led by location group g00175 (PK); its share of $R$ | 8 | 0.76 to 1.31 |
| *Inclusion criteria applied in order (item B4)* | | |
| Evaluable cells $\geq 400$ | 47 excluded | 952 remaining |
| Flooded cells $\geq 20$ | 452 excluded | 500 remaining |
| Flooded share $\leq 80$ % | 0 excluded | 500 remaining |
| Riverine (or untyped inland) share of the flooding $\geq 0.5$ | 0 excluded | 500 remaining |
| A settlement inside the grid | 0 excluded | 500 remaining |
| *Flood typing (item B13) and the GFM window (item B12)* | | |
| AOIs with untyped flooding; counted riverine by the inland rule | 150; 154 | with the ice-jam AOIs |
| AOIs with typed flooding of another kind; descriptions | 6 | Ice jam flood (12449); Landslide (268); None (2) |
| Flood traces as a share of the flooded area: optical; Sentinel-1; other radar | 14 %; 0.0 %; 0.0 % | |
| Distance of the AOI centroid from the coast: minimum; 10th percentile; median (km) | 1.4; 13.8; 79 | 81 AOIs within 20 km |
| AOIs with GFM images: with one on an EMS image day; only on adjacent days | 240; 77 | of 317 |
| *Labels near settlements and built-up land at 10 m (item B18)* | | |
| Odds ratio, ground 0.5–1 km against within 500 m: without; with the built-up share | 1.12 (1.05–1.20); 1.05 (0.98–1.13) | |
| Odds ratio, ground beyond 1 km against within 500 m: without; with the built-up share | 1.29 (1.12–1.48); 1.17 (1.02–1.34) | |
| Odds ratio, water-related names within 500 m (cluster-robust): without; with the built-up share | 1.01 (0.77–1.32); 0.98 (0.75–1.29) | |
| water-related, pairs bootstrap 95 %: without; with | 0.78–1.31; 0.75–1.28 | |
| *Every evaluable cell: where the flooded cells within 500 m of water-related names are (item B2)* | | |
| Cambodia | 1,299 (35 %) | 4 AOIs |
| Myanmar | 909 (24 %) | 11 AOIs |
| Bangladesh | 384 (10 %) | 2 AOIs |
| Pakistan | 264 (7.0 %) | 12 AOIs |
| Portugal | 87 (2.3 %) | 9 AOIs |
| Germany | 84 (2.2 %) | 31 AOIs |
| Activation EMSR052 (Cambodia) | 1,124 (30 %) | 3 AOIs |
| Activation EMSR130 (Myanmar/Burma) | 909 (24 %) | 11 AOIs |
| Odds ratio without Pakistan and Bangladesh (pairs 95 %) | 0.87 (0.73–1.07) | 3,105 flooded cells |
| *Penalty on the name features and the composition only (item B11)* | | |
| $R$; chosen $C$ for M1c and M3n (range over folds) | 0.004 | $10^{-7}$ to $10^{-4}$; $10^{-7}$ to $10^{-4}$; at the edge of the widened grid in 1 of 10 folds |

Table A27: Whether areas with more water-related names flood more, within language regions (exploratory; amendment 10, item B13). Regression of the empirical logit of each AOI's flooded fraction, $\log\{(k+0.5)/(n-k+0.5)\}$, on the share of its analysed cells whose nearest name is water-related and on the AOI mean of the baseline model's held-out logit, with activation fixed effects, for the AOIs of countries read with the German, English or French lexicon blocks; AOIs that are the only analysed AOI of their activation drop out with the fixed effects. Standard errors clustered by activation (CR1), 95 % intervals from the $t$ distribution with one degree of freedom fewer than the activations.

| Language | AOIs | Activations | Share water-related | Mean logit of M1 |
|---|---|---|---|---|
| German | 54 | 16 | −3.58 (−21.02 to +13.85) | +0.47 (+0.10 to +0.84) |
| English | 111 | 30 | −2.27 (−6.39 to +1.86) | +0.57 (+0.17 to +0.97) |
| French | 58 | 14 | −3.62 (−12.16 to +4.92) | +0.79 (−0.06 to +1.64) |
| All three | 212 | 56 | −2.38 (−6.04 to +1.28) | +0.56 (+0.29 to +0.84) |

Table A28: Post hoc runs and controls of amendment 6, primary split, corrected code. $R$ is the within-area increment as a fraction of the baseline gain of the same run, with the fixed-margin 90 % interval and the verdict of the decision table. At the T0 rung, the 95 % interval of $R$ is −0.024 to 0.002 with the corrected code against −0.017 to 0.001 with the frozen code (Table 2 of the paper), so a 90 % interval here can be wider than the 95 % interval there. The spread of the location groups' contributions grew by a factor of 1.48, almost all of it from two large AOIs, EMSR097 AOI01 (BD) and EMSR838 AOI05 (PK), whose contributions changed when the models were refitted; without them the factor is 0.87.

| Run | $R$ | 90 % interval | Verdict | Sample |
|---|---|---|---|---|
| Flood-label oracle (same event, leaks the outcome) | 0.513 | 0.338 to 0.715 | positive-meaningful | 500 AOIs |
| Recurrent surface water (JRC GSW) | 0.006 | −0.009 to 0.022 | equivalent-null | 500 AOIs |
| Recurrent surface water without permanent water | 0.004 | −0.020 to 0.027 | equivalent-null | 500 AOIs |
| Names on T0$^-$ (no river distances, no water land cover) | −0.008 | −0.019 to −0.001 | negative | 500 AOIs |
| Names on T0 (same run) | −0.008 | −0.021 to 0.001 | equivalent-null | 500 AOIs |
| Strict reading of the names | −0.003 | −0.006 to 0.0004 | equivalent-null | 500 AOIs |
| Language-model reading, primary split | −0.001 | −0.008 to 0.005 | equivalent-null | 500 AOIs |
| Gazetteer without city sections, historical places, duplicates | −0.003 | −0.007 to 0.0004 | equivalent-null | 500 AOIs |
| Without AOIs lacking a hydrography layer | −0.002 | −0.006 to 0.004 | equivalent-null | 285 AOIs |
| Without AOIs lacking a flood type | −0.009 | −0.019 to −0.001 | negative | 346 AOIs |
| Events held out whole: real increment among 200 reassignments | — | percentile 48 | — | — |
| Every AOI weighted equally | −0.0003 | −0.004 to 0.004 | equivalent | — |

## What assignment through settlements can carry

Table A29: Injected odds ratio (shape A: all ground nearest a water-related name) whose mean simulated gain reaches each interval's upper limit of $R$ minus the gain expected without an effect, under the four no-effect references (exploratory, amendment 13, item B2); in parentheses, the value on the scale of the estimator of Eq. (2) of the paper. A bound is not a test inversion, and the Monte Carlo ranges of the bound tables are not confidence intervals. References: recomputed composition (amendment 8), no reassignment, centred reassignments, and the reassignments of amendment 5.

| Interval | Upper $R$ | Recomputed | None | Centred | Amendment 5 |
|---|---|---|---|---|---|
| Protocol's interval (models fixed) | 0.003 | 1.25 (1.36) | 1.17 (1.24) | 1.28 (1.40) | 1.47 (1.69) |
| Location groups, second stage refitted (amendment 8) | 0.013 | 1.57 (1.83) | 1.50 (1.73) | 1.59 (1.86) | 1.76 (2.11) |
| Location groups, refitted, denominator resampled (amendment 11) | 0.008 | 1.40 (1.59) | 1.34 (1.50) | 1.43 (1.62) | 1.64 (1.94) |
| Activations × countries, refitted (amendment 11) | 0.036 | 2.11 (2.63) | 2.06 (2.55) | 2.13 (2.65) | 2.25 (2.83) |
| Spread over fold assignments (amendment 5) | 0.004 | 1.31 (1.45) | 1.23 (1.34) | 1.33 (1.47) | 1.53 (1.77) |
| Corrected test (a) | 0.004 | 1.29 (1.43) | 1.21 (1.31) | 1.31 (1.45) | 1.51 (1.75) |
| Corrected test (b) | 0.016 | 1.65 (1.95) | 1.58 (1.85) | 1.67 (1.97) | 1.82 (2.20) |
| Corrected test (c) | 0.031 | 1.99 (2.45) | 1.93 (2.37) | 2.00 (2.47) | 2.16 (2.70) |
| Corrected test (d) | 0.031 | 1.99 (2.45) | 1.93 (2.37) | 2.01 (2.48) | 2.16 (2.70) |
| *Effect within 1 km of the settlement (shape BK; amendment 12, item C1)* | | | | | |
| Margin | 0.050 | 7.55 | | | |
| Models fixed | | 2.13 | 1.82 | 2.20 | 2.92 |
| Second stage refitted | | 3.19 | 2.99 | 3.26 | 3.98 |

Table A30: Effects injected into the observed labels for all ground nearest a water-related name, with the base models fixed and with every model refitted on the injected labels (amendment 7; three repetitions with the injections of amendment 6): mean gain in ΔBS over the uninjected run ($\times 10^{-4}$), the same with fixed base models over the ten repetitions of amendment 6, the ratio of the refitted to the fixed gain, the baseline gain (M0 to M1) of the refitted models in the injected labels ($\times 10^{-4}$) and the share of held-out labels changed by the injection (%). Odds ratios at the margin and bounds are not read from this simulation: it has no effect below an odds ratio of 1.65, and a straight line from zero across the convex gain curve would understate them (amendment 8, addendum 9).

| Odds ratio | Fixed | Fixed, 10 rep. | Refitted | Ratio | Baseline gain | Labels changed |
|---|---|---|---|---|---|---|
| 1.65 | 1.36 | 1.37 | 1.23 | 0.91 | 78.9 | 0.15 |
| 2.72 | 5.12 | 5.18 | 4.88 | 0.95 | 78.9 | 0.34 |
| 4.48 | 12.38 | 12.39 | 11.66 | 0.94 | 78.1 | 0.55 |

Table A31: How an effect confined to the ground a name refers to appears in the odds ratio for all ground nearest the name (amendment 10, addendum 1, item 3; arithmetic, no data). $\varphi$: share of the nearest ground that the named feature occupies; $\mathrm{OR}_n$: odds ratio of flooding there. Rare-outcome approximation: risk ratio $1+\varphi(\mathrm{OR}_n-1)$ on the ground of truly water-related names, observed through names read with precision $p_1 = 0.76$ and a water-related share $p_0 = 0.18$ among other names, both measured with the codes of the model that compiled the lexicon (Table 9 of the paper and Sect. 3.6 of the paper; $p_0$ from the names not read as water-related in the sample of Wikipedia derivations).

| $\mathrm{OR}_n$ | $\varphi = 0.05$ | $\varphi = 0.1$ | $\varphi = 0.25$ | $\varphi = 0.5$ |
|---|---|---|---|---|
| 1.5 | 1.01 | 1.03 | 1.07 | 1.14 |
| 2 | 1.03 | 1.06 | 1.14 | 1.27 |
| 3 | 1.06 | 1.11 | 1.27 | 1.49 |
| 5 | 1.11 | 1.22 | 1.49 | 1.85 |
| *Without misreading ($p_1 = 1$, $p_0 = 0$)* | | | | |
| 1.5 | 1.02 | 1.05 | 1.12 | 1.25 |
| 2 | 1.05 | 1.10 | 1.25 | 1.50 |
| 3 | 1.10 | 1.20 | 1.50 | 2.00 |
| 5 | 1.20 | 1.40 | 2.00 | 3.00 |

Table A32: The margin and the bounds of Sect. 3.3 of the paper as injected (nominal) odds ratios and on the scale of the estimator of Eq. (2) of the paper (exploratory; amendment 11, item B3). The estimator was applied to labels with known injected odds ratios (Table S18); its mean odds ratio, relative to the uninjected labels, is interpolated on the log scale. Beyond the largest injected odds ratio (3) the estimator value is given as a lower limit. The no-effect reference is the mean increment of reassigned names with rebuilt composition. Calibration curve for all ground (nominal → estimator): 1.25 → 1.37, 1.5 → 1.73, 2 → 2.47, 3 → 3.94.

| Quantity | Injected odds ratio | Estimator odds ratio |
|---|---|---|
| Margin ($R = 0.05$), all ground nearest a water-related name | 2.33 | 2.94 |
| Upper bound, all ground, fitted models held fixed | 1.25 | 1.36 |
| Upper bound, within 500 m, fitted models held fixed | 6.25 | $>4.73$ |
| Upper bound, all ground, with the spread over fold assignments | 1.31 | 1.45 |
| Upper bound, within 500 m, with the spread over fold assignments | 7.77 | $>4.73$ |
| Upper bound, all ground, with the second stage refitted | 1.57 | 1.83 |
| Upper bound, within 500 m, with the second stage refitted | 18.37 | $>4.73$ |
| Upper bound, all ground, flips correlated within each settlement's ground | 1.27 | 1.39 |
| Upper bound, all ground, effects that differ between settlements (mean odds ratio) | 1.23 | 1.33 |

Table A33: Odds ratios of flooding that the margin and the observed upper limits correspond to under the synthetic labels of the protocol's power analysis (amendment 2): an effect on all ground nearest a water-related name, and on the ground within 500 m of such a settlement. In parentheses the range over Monte Carlo resamples of the simulated curve; >: beyond the largest simulated effect (odds ratio 4.95 for all ground, 20.1 within 500 m). These translations assume the label model of the power analysis; the simulations on the observed labels (Table A65) give lower values.

| Target of the translation | All ground nearest the name | Within 500 m |
|---|---|---|
| Margin ($R = 0.05$) | 4.12 (4.04–4.22) | $>20$ |
| Upper 90 % limit of the observed increment | 1.53 (1.44–1.61) | 6.94 (4.12–9.74) |
| Same, two-way bootstrap | 1.72 (1.63–1.83) | 9.52 (6.56–12.64) |
| Upper limit, confirmatory run | 1.53 | 6.94 |
| Upper limit, fold assignment 1 | 1.19 | 3.22 |
| Upper limit, fold assignment 2 | 2.20 | $>20$ |
| Upper limit, location groups formed with a 10 km buffer | 1.76 | 10.14 |
| Upper limit, events held out whole | 4.10 | $>20$ |

Table A34: Positive control without flood labels (amendment 6): within-AOI difference between settlements whose name is in each class and other settlements in the mean occurrence of surface water (percentage points of time), its mean seasonality (months) and the occurrence without permanent water, within 500 m of the settlement point (JRC Global Surface Water 1984–2021); 95 % cluster-bootstrap intervals over location groups.

| Names | Occurrence | Seasonality | Occurrence, non-permanent |
|---|---|---|---|
| water-related (any) | +1.05 (+0.34 to +2.00) | +0.12 (+0.04 to +0.23) | +0.60 (+0.13 to +1.25) |
| strict reading | +1.36 (+0.41 to +2.65) | +0.15 (+0.04 to +0.30) | +0.74 (+0.11 to +1.62) |
| water | +0.48 (-0.22 to +1.55) | +0.06 (-0.02 to +0.19) | +0.19 (-0.18 to +0.85) |
| wetland | -0.34 (-1.01 to +0.34) | -0.05 (-0.13 to +0.04) | -0.29 (-0.79 to -0.04) |
| riverine | +4.04 (+1.88 to +6.01) | +0.43 (+0.20 to +0.67) | +2.56 (+0.76 to +4.17) |
| crossing | +1.87 (+0.58 to +3.38) | +0.22 (+0.07 to +0.38) | +1.24 (+0.32 to +2.35) |
| lowland | +0.03 (-0.14 to +0.22) | +0.00 (-0.02 to +0.03) | -0.01 (-0.09 to +0.08) |
| highland | -0.47 (-0.79 to -0.14) | -0.06 (-0.11 to -0.02) | -0.28 (-0.47 to -0.10) |

# Flooding near settlements

Table A35: Further checks of the odds ratio of flooding within 500 m of water-related names (exploratory, amendment 14, items B11, B13, B14 and C1): the estimator of Eq. (2) of the paper on the capped EMS table (AOI fixed effects, distance bands, logit of the baseline model's held-out prediction, 95 % intervals clustered by location group), the planned GFM estimate with a jackknife interval, the uncapped estimate's bootstrap distribution and the check of the positional join.

| Check | Odds ratio (95 %) | Cells | Flooded near |
|---|---|---|---|
| Capped EMS table (Sect. 3.4 of the paper) | 1.01 (0.77–1.32) | 1,489,381 | 361 |
| Riverine cover as a fractional response | 0.98 (0.80–1.21) | 1,507,486 | 402 |
| Typed AOIs only | 1.00 (0.77–1.30) | 1,002,686 | 214 |
| With log(1 + population) and 10 m built-up share | 0.98 (0.75–1.29) | 1,489,381 | 361 |
| Planned GFM estimate, leave-one-group-out jackknife | 0.55 (0.36–0.84) | range without one group 0.51–0.63 | |
| Observable ground only (frozen reading) | 1.00 (0.76–1.33) | 950,840 | 359 |
| Observable ground only (edited lexicon) | 1.16 (0.82–1.64) | 950,840 | 261 |
| Uncapped EMS, edited lexicon: resamples of the pairs bootstrap | point 0.84, median 0.87, skewness of log OR 1.4, 16 % above one | | |
| Join of cells, features and land cover by row position | 500 AOIs, 0 with different row counts; 20.6 million distances recomputed, largest difference 0.004 m | | |

Table A36: Diagnosis of the GFM labels (exploratory, amendment 13, items B9(e) and B14): flooded shares by AOI on the cells that GFM can assess, the baseline gain when every model is trained and scored on GFM labels, and the agreement of the two label sets by distance from the nearest settlement and by built-up land (ESA WorldCover, 10 m; more than two-thirds of these cells contain none, so the cells are split into those with and without it).

| Quantity | Value |
|---|---|
| AOIs on both label sets | 192 |
| Median flooded share, EMS / GFM | 5.3 % / 4.7 % |
| AOIs where GFM mapped less flooding than EMS | 53 % |
| Correlation of the AOIs' flooded shares | 0.81 |
| Baseline gain on GFM labels (pooled) | $-6.6 \times 10^{-4}$ |
| Baseline gain on EMS labels (primary) | $7.6 \times 10^{-3}$ |
| AOIs with a negative baseline gain on GFM | 28 % |
| EMS and GFM, within 500 m of the settlement | both 6.3 %, EMS only 5.6 %, GFM only 1.0 %; GFM mapped 53 % of the EMS flooding (30,376 cells) |
| EMS and GFM, 500 m to 1 km | both 7.8 %, EMS only 5.2 %, GFM only 1.5 %; GFM mapped 60 % of the EMS flooding (59,234 cells) |
| EMS and GFM, beyond 1 km | both 6.8 %, EMS only 4.9 %, GFM only 1.9 %; GFM mapped 58 % of the EMS flooding (145,093 cells) |
| EMS and GFM, cells without built-up land | both 8.2 %, EMS only 5.3 %, GFM only 1.9 %; GFM mapped 61 % of the EMS flooding (187,766 cells) |
| EMS and GFM, cells with built-up land | both 2.2 %, EMS only 4.1 %, GFM only 0.7 %; GFM mapped 35 % of the EMS flooding (46,937 cells) |

Table A37: Positive control without flood labels (binary measures): within-AOI differences between cells (or settlements) whose nearest settlement name is in the class and other cells (or settlements) of the same AOI, with 95 % cluster-bootstrap intervals over location groups. "HAND $\leq$ 2 m" is whether any 25 m pixel of the 250 m cell lies within 2 m of the nearest drainage (the cell's minimum HAND); it holds for most cells, so the shares in each group are given. Lower half: the cell lies in the lower half of its AOI by mean elevation. Continuous measures are in Table A38. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

| Nearest name | Scale | HAND $\leq$ 2 m in cell (pp) | Class/other (%) | Lower half (pp) | AOIs |
|---|---|---|---|---|---|
| water-related (any) | all cells | +0.2 [-1.0, +1.3] | 82/82 | +0.6 [-2.9, +4.1] | 310 |
| | at settlement[a] | +5.7 [+3.5, +8.1] | 89/83 | +5.0 [+1.5, +8.7] | 151 |
| water | all cells | -0.7 [-2.0, +0.4] | 83/83 | -4.0 [-8.9, +0.7] | 176 |
| | at settlement[a] | +3.2 [+0.1, +6.6] | 88/84 | -3.7 [-9.7, +2.7] | 64 |
| wetland | all cells | +0.4 [-3.3, +3.8] | 85/84 | +1.5 [-7.4, +11.3] | 41 |
| | at settlement[a] | +6.7 [-4.3, +22.1] | 88/81 | -1.5 [-18.9, +25.0] | 12 |
| riverine | all cells | +2.1 [+0.1, +4.4] | 90/87 | +12.8 [+5.1, +20.3] | 95 |
| | at settlement[a] | +4.7 [+0.5, +10.4] | 92/87 | +16.4 [+8.9, +24.5] | 38 |
| crossing | all cells | +2.1 [+0.6, +3.6] | 81/79 | +9.4 [+4.3, +14.5] | 102 |
| | at settlement[a] | +7.5 [+0.5, +14.5] | 91/84 | +9.3 [+0.8, +17.6] | 34 |
| lowland | all cells | -0.9 [-3.2, +1.2] | 78/79 | -4.9 [-10.4, +0.9] | 160 |
| | at settlement[a] | +5.5 [+0.4, +10.7] | 85/79 | +3.4 [-3.7, +9.8] | 60 |
| highland | all cells | -3.1 [-4.5, -1.6] | 78/81 | -7.7 [-11.7, -3.5] | 253 |
| | at settlement[a] | -10.1 [-13.4, -6.8] | 72/82 | -8.9 [-13.0, -4.9] | 137 |
| water-related (any) | at settlement, coordinates not rounded[b] | +6.1 [+3.7, +8.4] | 90/83 | +4.9 [+1.3, +8.7] | 135 |
| highland | at settlement, coordinates not rounded[b] | -10.4 [-13.9, -6.8] | 71/82 | -9.0 [-13.2, -4.9] | 122 |

Table A38: Positive control without flood labels (continuous measures), as in Table A37: differences in log(1 + distance to the nearest mapped river), in the cell's mean height above nearest drainage and in its HAND percentile within the AOI. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

| Nearest name | Scale | log(1 + river distance) | Mean HAND (m) | HAND percentile (pp) |
|---|---|---|---|---|
| water-related (any) | all cells | -0.04 [-0.09, +0.01] | +0.7 [-0.7, +2.3] | -0.04 [-1.6, +1.7] |
| | at settlement[a] | -0.22 [-0.33, -0.11] | -2.0 [-2.9, -1.3] | -4.6 [-6.1, -3.1] |
| water | all cells | +0.01 [-0.07, +0.08] | +1.1 [-0.3, +2.8] | +2.0 [-0.1, +4.4] |
| | at settlement[a] | -0.17 [-0.33, -0.01] | -0.6 [-1.7, +0.5] | -2.0 [-4.3, +0.4] |
| wetland | all cells | -0.001 [-0.14, +0.14] | +0.1 [-2.8, +3.7] | -3.5 [-8.2, +1.6] |
| | at settlement[a] | +0.17 [-0.28, +0.50] | -0.7 [-5.7, +3.9] | -1.7 [-13.6, +5.3] |
| riverine | all cells | -0.01 [-0.15, +0.16] | -0.6 [-3.1, +2.2] | -5.0 [-8.7, -1.3] |
| | at settlement[a] | -0.12 [-0.42, +0.14] | -2.2 [-4.1, -0.7] | -7.7 [-11.9, -4.4] |
| crossing | all cells | -0.18 [-0.27, -0.10] | -1.0 [-3.1, +0.8] | -1.5 [-4.2, +0.9] |
| | at settlement[a] | -0.54 [-0.86, -0.28] | -2.6 [-4.5, -1.0] | -5.3 [-9.9, -1.5] |
| lowland | all cells | -0.02 [-0.11, +0.07] | +1.7 [-1.0, +5.0] | +1.1 [-1.6, +3.9] |
| | at settlement[a] | -0.22 [-0.40, -0.05] | -2.3 [-3.9, -0.9] | -3.2 [-6.7, +0.1] |
| highland | all cells | +0.08 [+0.03, +0.13] | +3.4 [+1.5, +5.9] | +3.8 [+2.1, +5.5] |
| | at settlement[a] | +0.11 [+0.01, +0.23] | +5.3 [+3.8, +7.1] | +8.5 [+6.0, +10.8] |
| water-related (any) | at settlement, coordinates not rounded[b] | -0.23 [-0.35, -0.11] | -2.1 [-2.9, -1.5] | -5.0 [-6.7, -3.5] |
| highland | at settlement, coordinates not rounded[b] | +0.11 [-0.01, +0.22] | +5.3 [+3.7, +7.1] | +8.8 [+6.3, +11.3] |

Table A39: Where settlements with water-related names stand (exploratory, amendment 8; no flood labels). Within-AOI difference between settlements whose name is in the class and other settlements of the same AOI in the height of the settlement's cell above the nearest point of a river draining at least 100 or 1000 km$^2$ (HydroRIVERS; mean elevation of the settlement's cell minus the lowest elevation of the river's cell, m), and in the logarithm of the drainage area of the nearest river; 95 % cluster-bootstrap intervals over location groups. Last row: the HAND of the settlement's site, for names that contain the name of a river, stream or lake within 5 km in the gazetteer.

| Names | Above rivers ≥100 km$^2$ | ≥1000 km$^2$ | log drainage area | Settlements |
|---|---|---|---|---|
| Water-related (pooled) | −5.0 (−9.0 to −1.3) | −4.8 (−9.0 to −1.0) | 0.11 (−0.02 to 0.24) | 11120 |
| Floodplain reading (riverine, wetland) | −4.9 (−10.6 to −0.2) | −3.2 (−9.6 to 2.6) | 0.28 (−0.09 to 0.67) | 5150 |
| Water bodies | 0.9 (−4.5 to 6.6) | −0.9 (−6.8 to 5.1) | −0.21 (−0.48 to 0.05) | 6156 |
| Wetland | 0.1 (−17.3 to 13.5) | 5.3 (−17.4 to 28.2) | 0.18 (−0.90 to 1.39) | 1478 |
| Riverine ground | −4.4 (−12.0 to 2.3) | −5.7 (−13.1 to 0.9) | 0.31 (−0.07 to 0.71) | 4077 |
| Crossings | −8.7 (−13.9 to −3.9) | −7.5 (−13.2 to −3.0) | 0.61 (0.18 to 1.13) | 3121 |
| Lowland | −6.0 (−17.3 to 4.7) | −4.3 (−16.8 to 7.4) | 0.21 (−0.03 to 0.43) | 5565 |
| Names containing a nearby river or lake name (HAND of the site) | −2.4 (−4.4 to −0.6) | — | — | 3931 |

Table A40: Within-AOI odds ratios of flooding for cells whose nearest settlement name is in the class, from logistic regressions with AOI fixed effects, unadjusted or with the logit of the held-out terrain-model prediction (M1) as a covariate; 95 % intervals from cluster-robust (location-group) standard errors; 479 AOIs (AOIs whose sampled cells are all dry carry no information and are omitted). [r] All cells, one indicator per class: the analysis planned in amendment 2. The distance-banded models (class × band, with band main effects) were not planned. The distance-band row gives the odds of a flood label 0.5–1 km and more than 1 km from the nearest settlement relative to within 500 m. Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 % excluded. A cluster bootstrap over location groups (200 replicates, AOIs re-entered with their own intercepts) gives similar intervals, e.g. 0.73–1.28 for water-related names within 500 m given the baseline model.

| Nearest name | All cells[r] | | Unadjusted, by distance | | | Given the baseline model, by distance | | |
|---|---|---|---|---|---|---|---|---|
| | unadjusted | given terrain | ≤ 500 m | 0.5–1 km | > 1 km | ≤ 500 m | 0.5–1 km | > 1 km |
| water-related (any) | 1.00 [0.81, 1.23] | 1.02 [0.83, 1.25] | 1.05 [0.80, 1.36] | 1.11 [0.91, 1.34] | 0.95 [0.74, 1.22] | 1.01 [0.77, 1.32] | 1.06 [0.85, 1.33] | 1.00 [0.80, 1.26] |
| water | 0.85 [0.60, 1.21] | 0.97 [0.73, 1.29] | 0.89 [0.67, 1.20] | 0.86 [0.67, 1.11] | 0.85 [0.52, 1.38] | 0.77 [0.57, 1.05] | 0.80 [0.62, 1.03] | 1.07 [0.74, 1.55] |
| wetland | 0.98 [0.46, 2.09] | 1.09 [0.47, 2.53] | 1.19 [0.52, 2.73] | 1.10 [0.45, 2.69] | 0.92 [0.45, 1.84] | 1.60 [0.71, 3.59] | 1.32 [0.48, 3.62] | 0.80 [0.38, 1.67] |
| riverine | 1.30 [0.91, 1.86] | 1.17 [0.74, 1.83] | 1.46 [0.94, 2.26] | 1.48 [1.03, 2.11] | 1.16 [0.80, 1.69] | 1.41 [0.92, 2.16] | 1.38 [0.92, 2.08] | 1.04 [0.64, 1.70] |
| crossing | 1.02 [0.72, 1.44] | 0.90 [0.65, 1.26] | 0.75 [0.42, 1.35] | 1.01 [0.64, 1.60] | 1.04 [0.71, 1.53] | 0.77 [0.46, 1.26] | 0.89 [0.58, 1.35] | 0.92 [0.63, 1.35] |
| lowland | 0.83 [0.56, 1.23] | 0.93 [0.65, 1.33] | 0.80 [0.42, 1.54] | 1.06 [0.61, 1.86] | 0.76 [0.49, 1.18] | 0.87 [0.50, 1.50] | 1.16 [0.68, 1.97] | 0.87 [0.59, 1.29] |
| highland | 1.00 [0.83, 1.22] | 1.08 [0.90, 1.30] | 0.98 [0.74, 1.29] | 1.04 [0.80, 1.35] | 1.00 [0.81, 1.24] | 1.21 [0.88, 1.68] | 1.10 [0.89, 1.36] | 1.07 [0.86, 1.33] |
| Distance band (all names) | — | — | 1 (reference) | 1.61 [1.48, 1.76] | 1.97 [1.64, 2.37] | 1 (reference) | 1.12 [1.05, 1.20] | 1.29 [1.12, 1.48] |
| water-related, built-up and tree cover removed | — | — | 1.03 [0.79, 1.36] | 1.09 [0.90, 1.34] | 0.94 [0.73, 1.22] | 1.03 [0.77, 1.37] | 1.09 [0.86, 1.37] | 1.01 [0.79, 1.29] |

Table A41: Odds ratios of flooding near settlements by class and distance band on the EMS labels, given the baseline model's prediction (AOI fixed effects), with cluster-robust and pairs cluster bootstrap 95 % intervals over location groups (2,000 replicates; amendment 7). The two blocks are separate models (water-related names; the six classes).

| Names, distance | Odds ratio | Cluster-robust | Pairs bootstrap |
|---|---|---|---|
| water-related, ≤500 m | 1.01 | 0.77–1.32 | 0.77–1.31 |
| water-related, 0.5–1 km | 1.06 | 0.85–1.33 | 0.84–1.30 |
| water-related, >1 km | 1.00 | 0.80–1.26 | 0.79–1.25 |
| water, ≤500 m | 0.77 | 0.57–1.05 | 0.54–1.03 |
| water, 0.5–1 km | 0.80 | 0.62–1.03 | 0.60–1.02 |
| water, >1 km | 1.07 | 0.74–1.55 | 0.71–1.49 |
| wetland, ≤500 m | 1.60 | 0.71–3.59 | 0.41–3.62 |
| wetland, 0.5–1 km | 1.32 | 0.48–3.62 | 0.32–3.06 |
| wetland, >1 km | 0.80 | 0.38–1.67 | 0.35–1.61 |
| riverine, ≤500 m | 1.41 | 0.92–2.16 | 0.81–2.19 |
| riverine, 0.5–1 km | 1.38 | 0.92–2.08 | 0.84–2.04 |
| riverine, >1 km | 1.04 | 0.64–1.70 | 0.60–1.58 |
| crossing, ≤500 m | 0.77 | 0.46–1.26 | 0.42–1.22 |
| crossing, 0.5–1 km | 0.89 | 0.58–1.35 | 0.56–1.32 |
| crossing, >1 km | 0.92 | 0.63–1.35 | 0.59–1.29 |
| lowland, ≤500 m | 0.87 | 0.50–1.50 | 0.45–1.41 |
| lowland, 0.5–1 km | 1.16 | 0.68–1.97 | 0.61–1.83 |
| lowland, >1 km | 0.87 | 0.59–1.29 | 0.57–1.23 |
| highland, ≤500 m | 1.21 | 0.88–1.68 | 0.83–1.64 |
| highland, 0.5–1 km | 1.10 | 0.89–1.36 | 0.85–1.34 |
| highland, >1 km | 1.07 | 0.86–1.33 | 0.85–1.35 |

Table A42: The names increment within areas (M1c→M3n) by class of the nearest name, on the Brier scale (post hoc; planned in amendment 2 as descriptive quantities, the 95 % intervals and the difference between real and permuted names in its fourth addendum). All Brier-score differences are in units of $10^{-4}$; positive values mean that names improved held-out predictions. Intervals come from the cluster bootstrap over location groups restricted to the cells of each class (classes overlap: a name can belong to several). Baseline ΔBS is the baseline gain (M0→M1) in the same cells, the denominator of the class verdicts in Table A18; the pooled equivalence margin is 3.78. [a]For all cells, the difference between real and permuted names is given as a ratio in Table S6.

| Nearest name | Cells | Units | ΔBS | 90 % interval | 95 % interval | Real − permuted (95 %) | Baseline ΔBS |
|---|---|---|---|---|---|---|---|
| all cells (primary) | 1,507,486 | 195 | −0.264 | [−0.826, +0.210] | [−0.956, +0.279] | —[a] | 75.5 |
| water-related (any) | 89,404 | 146 | −0.448 | [−2.87, +1.98] | [−3.30, +2.39] | +0.895 [−3.16, +5.23] | 125 |
| water | 35,814 | 118 | +2.84 | [−0.263, +6.55] | [−0.777, +7.51] | +6.33 [+0.124, +15.2] | 265 |
| wetland | 2,857 | 51 | −7.68 | [−19.4, −0.00298] | [−22.6, +0.420] | −10.6 [−24.7, −1.26] | 15.4 |
| riverine | 13,841 | 85 | −14.2 | [−29.6, +0.259] | [−33.2, +2.93] | −11.0 [−31.1, +8.00] | 58.1 |
| crossing | 13,918 | 83 | +2.08 | [−2.65, +7.77] | [−3.61, +9.03] | +0.585 [−5.71, +8.62] | 19.7 |
| lowland | 24,552 | 101 | +2.01 | [−2.43, +7.18] | [−3.26, +8.38] | +1.15 [−4.03, +6.98] | 27.1 |
| highland | 58,021 | 130 | −1.10 | [−3.26, +1.12] | [−3.63, +1.54] | −0.761 [−2.79, +1.45] | 86.2 |

Table A43: Within-AOI flood contrasts by distance from the cell to its nearest settlement (not planned; added while the planned analyses were written, with their full reporting planned in amendment 2, addendum 2): the share of cells labelled flooded nearest a water-related name minus the share nearest other names, in percentage points, observed and net of the held-out predictions of the baseline model with the regional composition of names (M1c), with 95 % cluster-bootstrap intervals over location groups. AOIs enter a band when both groups have at least 20 cells in it; they are averaged with weights equal to the size of the smaller group, or scaled to the uncapped number of evaluable cells of each AOI, which lets a few large AOIs dominate.

| Distance to the nearest settlement | AOIs | Location groups | Weights: smaller group | | Weights: uncapped AOI size | |
|---|---|---|---|---|---|---|
| | | | observed | net of terrain | observed | net of terrain |
| ≤ 250 m | 37 | 31 | +1.1 [+0.04, +2.9] | +0.5 [−0.3, +1.8] | +0.1 [−1.0, +2.4] | −0.8 [−2.3, +1.6] |
| 250–500 m | 98 | 70 | +0.2 [−0.6, +1.2] | −0.1 [−0.8, +0.9] | −1.1 [−2.6, +0.9] | −2.1 [−4.3, +0.6] |
| 0.5–1 km | 185 | 111 | +0.4 [−0.4, +1.4] | +0.2 [−0.5, +1.1] | +0.8 [−1.7, +3.9] | −0.04 [−2.8, +2.9] |
| > 1 km | 263 | 134 | −0.2 [−1.1, +0.7] | −0.1 [−0.9, +0.6] | +2.9 [−0.4, +5.5] | +1.6 [−0.7, +3.4] |

Table A44: Odds ratios of flooding within 500 m of settlements, given the held-out terrain prediction (AOI fixed effects, distance bands, logit of the baseline model's prediction; exploratory, amendment 8, all not planned in the protocol). Pairs: cluster bootstrap over location groups with the regression refitted, 2,000 replicates. Cleaned reading: without two-letter affixes and the English suffixes that also read transferred names (0.6 % of the cells change their reading). Uncapped: every evaluable cell of the analysed AOIs, with the predictions of the uncapped sensitivity run. Misclassification: probabilistic bias analysis with the rates measured against recorded etymologies by the model that compiled the lexicon. [s]AOIs that enter the model, which needs flooded and dry cells in an AOI, of the AOIs whose EMS map was not drawn from Sentinel-1 (capped sample; other tables count the AOIs of their own samples). Columns 4–6: flooded cells within 500 m of water-related names, location groups with any cell within 500 m of a water-related name (flooded or not), and the AOIs that enter each model (AOIs with flooded and dry cells). The uncapped row uses the readings of the frozen code; the corrected readings are in the table of uncapped estimates of amendment 11.

| Estimate | Odds ratio | 95 % interval | Flooded | Groups | AOIs |
|---|---|---|---|---|---|
| *Pooled water-related class, EMS labels (pairs cluster bootstrap)* | | | | | |
| All AOIs | 1.01 | 0.77–1.31 | 361 | 134 | 479 |
| AOIs with a recorded flood type | 1.00 | 0.77–1.30 | 214 | 91 | 331 |
| Without AOIs lacking a hydrography layer | 0.85 | 0.62–1.19 | 183 | 84 | 269 |
| Without Oceania and the Americas | 0.96 | 0.71–1.29 | 303 | 107 | 402 |
| Without Australia | 1.00 | 0.74–1.30 | 322 | 124 | 436 |
| Cleaned reading | 1.04 | 0.76–1.38 | 271 | 140 | 479 |
| Every evaluable cell (no cap; frozen-code readings) | 0.91 | 0.76–1.15 | 3,749 | 137 | 500 |
| Without AOIs in Pakistan | 1.09 | 0.77–1.45 | 271 | 127 | 467 |
| Without AOIs in Pakistan and Bangladesh | 0.97 | 0.72–1.24 | 184 | 125 | 465 |
| Truly water-related names (misclassification corrected) | 1.00 | 0.63–1.63 | | | |
| *By the river nearest the settlement, and adjusted for its height (cluster-robust)* | | | | | |
| Draining less than 100 $km^2$ | 0.91 | 0.58–1.42 | 188 | 117 | 431 |
| Draining 100–1000 $km^2$ | 1.61 | 0.98–2.64 | 87 | 93 | 351 |
| Draining at least 1000 $km^2$ | 0.84 | 0.52–1.36 | 86 | 89 | 368 |
| Cells with a height above the main river, unadjusted | 1.00 | 0.75–1.33 | 331 | 118 | 427 |
| adjusted for the height above the river | 1.00 | 0.75–1.33 | 331 | 118 | 427 |
| Names containing a nearby river or lake name | 0.88 | 0.58–1.33 | 66 | 112 | 479 |
| *By class, all classes in one model (Bonferroni 95 % interval; Holm-adjusted p in the last column)* | | | | | |
| Water bodies | 0.77 | 0.51–1.17 | | | 0.58 |
| Wetland | 1.60 | 0.54–4.76 | | | 0.94 |
| Riverine ground | 1.41 | 0.80–2.50 | | | 0.58 |
| Crossings | 0.77 | 0.39–1.50 | | | 0.94 |
| Lowland | 0.87 | 0.42–1.81 | | | 0.94 |
| Highland | 1.21 | 0.79–1.87 | | | 0.94 |
| *Riverine flood cover as a fractional outcome, water-related class by distance band* | | | | | |
| Within 500 m | 0.98 | 0.80–1.21 | | | |
| 500 m–1 km | 1.05 | 0.86–1.27 | | | |
| Beyond 1 km | 1.01 | 0.83–1.23 | | | |
| *The two label sets on the same cells (pairs bootstrap; jackknife with t quantiles)* | | | | | |
| EMS | 0.78 | 0.60–1.07 | 80 groups, 176 AOIs; jackknife 0.54–1.11 | | |
| GFM | 0.55 | 0.34–0.81 | 80 groups, 176 AOIs; jackknife 0.36–0.85 | | |
| Ratio of the odds ratios, GFM over EMS | 0.71 | 0.47–0.92 | | | |
| Floodplain reading, EMS | 0.83 | 0.38–1.81 | 14 | 31 | 182 |
| Floodplain reading, GFM | 0.31 | 0.09–1.00 | 5 | 31 | 182 |
| EMS map not from Sentinel-1: GFM | 0.48 | 0.24–0.98 | | | 63 of 68[s] |
| EMS map not from Sentinel-1: EMS | 0.77 | 0.46–1.28 | | | 66 of 68[s] |

Table A45: Where the flooded cells within 500 m of settlements with water-related names lie (EMS labels; descriptive): 361 flooded cells among 10,048 such cells, in the AOIs of the odds ratios near settlements (AOIs with both flooded and dry cells). Share: of all flooded cells near water-related names. For the AOIs in Pakistan the fourth column gives the flooded cells of all evaluable cells (amendment 8, addendum 6). AOIs: the AOIs with such cells whose nearest settlement lies in the country or activation; an AOI whose cells are nearest settlements in several countries counts in each, so the column sums to more than the number of AOIs (510 by country).

| Country or activation | Flooded | Cells | Share | AOIs |
|---|---|---|---|---|
| *By country of the nearest settlement* | | | | |
| PK | 90 | 683 | 25 % | 12 |
| BD | 87 | 453 | 24 % | 2 |
| AU | 39 | 728 | 11 % | 43 |
| DE | 37 | 1,888 | 10 % | 29 |
| PT | 23 | 315 | 6.4 % | 9 |
| GB | 15 | 1,807 | 4.2 % | 18 |
| MX | 11 | 224 | 3.0 % | 3 |
| SI | 8 | 180 | 2.2 % | 6 |
| VN | 6 | 278 | 1.7 % | 16 |
| ES | 6 | 142 | 1.7 % | 25 |
| MM | 5 | 192 | 1.4 % | 11 |
| IR | 4 | 152 | 1.1 % | 3 |
| RO | 3 | 227 | 0.8 % | 12 |
| HR | 3 | 101 | 0.8 % | 27 |
| KH | 3 | 37 | 0.8 % | 4 |
| Other 30 countries | 21 | 2,641 | 5.8 % | 290 |
| *The activations with the most such cells* | | | | |
| EMSR097: Floods in Bangladesh (2014) | 87 | 453 | 24 % | 2 |
| EMSR629: Floods in Pakistan (2022) | 73 | 149 | 20 % | 3 |
| EMSR047: Flood in Sachsen-Anhalt- Germany (2013) | 23 | 199 | 6.4 % | 2 |
| EMSR567: Floods in Queensland- Australia (2022) | 22 | 159 | 6.1 % | 8 |
| EMSR864: Flood in Portugal and Spain (2026) | 22 | 124 | 6.1 % | 5 |
| EMSR838: Flood in Pakistan (2025) | 17 | 525 | 4.7 % | 7 |
| EMSR570: Flood in New South Wales- Australia (2022) | 14 | 47 | 3.9 % | 2 |
| EMSR845: Flood in Mexico (2025) | 9 | 88 | 2.5 % | 1 |
| EMSR150: Flood in England (2015) | 6 | 470 | 1.7 % | 4 |
| EMSR388: Flood in the Southeast of Spain (2019) | 6 | 62 | 1.7 % | 1 |
| *AOIs in Pakistan, by activation (evaluable cells; flooded near / near)* | | | | |
| EMSR838: Flood in Pakistan (2025) | 17 | 525 | 6,047 of 38,140 | 7 |
| EMSR629: Floods in Pakistan (2022) | 73 | 149 | 12,068 of 20,000 | 3 |
| EMSR631: Flood in Pakistan (2022) | 0 | 9 | 1,165 of 3,122 | 2 |

Table A46: The pooled odds ratio of flooding within 500 m of water-related names (EMS labels, the main model of amendment 8) with each activation that has flooded cells near water-related names left out in turn (exploratory; amendment 9; 48 activations). Flooded: flooded cells near water-related names in the activation left out. Intervals: 95 %, cluster-robust over location groups.

| Left out | Flooded | OR | 95 % | Left out | Flooded | OR | 95 % |
|---|---|---|---|---|---|---|---|
| EMSR044 | 2 | 1.01 | 0.77–1.33 | EMSR427 | 2 | 1.00 | 0.77–1.31 |
| EMSR047 | 23 | 1.00 | 0.76–1.32 | EMSR429 | 2 | 1.01 | 0.77–1.32 |
| EMSR052 | 2 | 1.02 | 0.78–1.33 | EMSR514 | 3 | 1.01 | 0.77–1.32 |
| EMSR055 | 1 | 1.01 | 0.77–1.32 | EMSR517 | 2 | 1.01 | 0.77–1.32 |
| EMSR069 | 4 | 1.03 | 0.78–1.35 | EMSR567 | 22 | 0.98 | 0.74–1.28 |
| EMSR070 | 4 | 1.01 | 0.77–1.32 | EMSR570 | 14 | 1.01 | 0.77–1.34 |
| EMSR073 | 1 | 1.01 | 0.77–1.32 | EMSR574 | 3 | 1.01 | 0.77–1.32 |
| EMSR097 | 87 | 0.91 | 0.72–1.14 | EMSR629 | 73 | 1.08 | 0.81–1.44 |
| EMSR107 | 4 | 1.00 | 0.76–1.31 | EMSR664 | 1 | 1.01 | 0.77–1.32 |
| EMSR130 | 5 | 1.04 | 0.80–1.35 | EMSR708 | 1 | 1.01 | 0.77–1.32 |
| EMSR142 | 1 | 1.01 | 0.77–1.32 | EMSR712 | 4 | 1.01 | 0.77–1.32 |
| EMSR147 | 3 | 1.01 | 0.77–1.33 | EMSR722 | 5 | 1.01 | 0.77–1.32 |
| EMSR150 | 6 | 1.00 | 0.76–1.31 | EMSR754 | 3 | 1.02 | 0.78–1.34 |
| EMSR167 | 1 | 1.01 | 0.77–1.32 | EMSR761 | 1 | 1.01 | 0.77–1.32 |
| EMSR199 | 1 | 1.00 | 0.77–1.32 | EMSR774 | 2 | 1.01 | 0.77–1.32 |
| EMSR203 | 2 | 1.01 | 0.77–1.32 | EMSR799 | 2 | 1.01 | 0.77–1.32 |
| EMSR258 | 1 | 1.01 | 0.77–1.32 | EMSR802 | 2 | 1.01 | 0.77–1.32 |
| EMSR275 | 1 | 1.01 | 0.77–1.32 | EMSR838 | 17 | 1.01 | 0.76–1.34 |
| EMSR293 | 1 | 1.01 | 0.77–1.32 | EMSR845 | 9 | 0.98 | 0.75–1.29 |
| EMSR352 | 1 | 1.02 | 0.78–1.33 | EMSR848 | 3 | 1.02 | 0.78–1.34 |
| EMSR388 | 6 | 0.99 | 0.76–1.30 | EMSR857 | 1 | 1.01 | 0.77–1.32 |
| EMSR407 | 2 | 1.00 | 0.77–1.31 | EMSR861 | 1 | 1.01 | 0.77–1.32 |
| EMSR419 | 3 | 1.00 | 0.77–1.32 | EMSR864 | 22 | 0.97 | 0.74–1.28 |
| EMSR424 | 3 | 1.01 | 0.77–1.33 | EMSR866 | 1 | 1.01 | 0.77–1.32 |

Table A47: The six class coefficients of the model with all lexicon classes within 500 m (EMS labels; the figure of odds ratios near settlements in the paper): unadjusted $p$, Holm-adjusted $p$ (Holm, 1979) and Benjamini–Hochberg $q$ (Benjamini and Hochberg, 1995) (exploratory; amendment 9).

| Class | $p$ | Holm $p$ | BH $q$ |
|---|---|---|---|
| Water bodies | 0.097 | 0.584 | 0.339 |
| Wetland | 0.258 | 0.942 | 0.356 |
| Riverine ground | 0.113 | 0.584 | 0.339 |
| Crossings | 0.297 | 0.942 | 0.356 |
| Lowland | 0.609 | 0.942 | 0.609 |
| Highland | 0.236 | 0.942 | 0.356 |

Table A48: Odds ratio near water-related names corrected for misreading over a grid of rates (exploratory, amendment 10, item B5): $\mathrm{OR} = (\mathrm{OR}^*(1-p_0)-(1-p_1))/(p_1-\mathrm{OR}^*p_0)$, with $\mathrm{OR}^*$ the observed estimate within 500 m and the limits of its pairs-bootstrap 95 % interval; $p_1$: share of water-related names among those read as water-related; $p_0$: share among the others. Undefined: the observed value cannot arise from these rates.

| $p_1$ | $p_0$ | Observed point | Observed lower | Observed upper |
|---|---|---|---|---|
| 0.6 | 0.05 | 1.02 | 0.60 | 1.57 |
| 0.6 | 0.10 | 1.02 | 0.57 | 1.64 |
| 0.6 | 0.20 | 1.02 | 0.49 | 1.89 |
| 0.6 | 0.30 | 1.03 | 0.39 | 2.45 |
| 0.7 | 0.05 | 1.01 | 0.66 | 1.48 |
| 0.7 | 0.10 | 1.02 | 0.64 | 1.53 |
| 0.7 | 0.20 | 1.02 | 0.59 | 1.69 |
| 0.7 | 0.30 | 1.02 | 0.52 | 1.98 |
| 0.8 | 0.05 | 1.01 | 0.70 | 1.41 |
| 0.8 | 0.10 | 1.01 | 0.69 | 1.45 |
| 0.8 | 0.20 | 1.02 | 0.65 | 1.56 |
| 0.8 | 0.30 | 1.02 | 0.60 | 1.74 |
| 0.9 | 0.05 | 1.01 | 0.74 | 1.36 |
| 0.9 | 0.10 | 1.01 | 0.73 | 1.39 |
| 0.9 | 0.20 | 1.01 | 0.70 | 1.47 |
| 0.9 | 0.30 | 1.02 | 0.66 | 1.59 |

Table A49: Odds ratio of flooding within 500 m of settlements with water-related names against other names of the same AOI, given the held-out prediction of the baseline model (exploratory; amendment 11, items B2 and B5): AOI fixed effects, distance bands and the logit of the prediction, with a 95 % pairs cluster bootstrap over location groups (2,000 replicates). Flooded: flooded cells within 500 m of water-related names and the location groups that hold them. MDE: the smallest odds ratios below and above one detectable with 80 % power (two-sided 5 %). The uncapped sample conditions on the predictions of the uncapped sensitivity run (the baseline model does not use names). The GFM labels are not independent of EMS where the EMS map was drawn from Sentinel-1. The number of AOIs not mapped from Sentinel-1 differs between tables and figures because each counts the AOIs that enter its own sample: uncapped or capped, cells that GFM can assess, and AOIs with both flooded and dry cells.

| Sample and labels | Odds ratio | 95 % interval | Flooded | Groups | MDE |
|---|---|---|---|---|---|
| *Every evaluable cell (no cap), corrected-code readings* | | | | | |
| EMS labels, 500 AOIs | 0.91 | 0.76–1.16 | 3,753 | 75 | 0.74, 1.35 |
| GFM labels, cells that GFM can assess | 0.82 | 0.60–1.07 | 434 | 29 | 0.66, 1.50 |
| EMS labels on the same cells | 0.89 | 0.67–1.16 | 560 | 36 | 0.68, 1.47 |
| GFM labels, AOIs not mapped from Sentinel-1 (111) | 0.85 | 0.43–1.46 | 166 | 15 | 0.44, 2.29 |
| EMS labels, the same AOIs and cells | 1.00 | 0.71–1.58 | 263 | 20 | 0.56, 1.79 |
| EMS labels, flooded fraction over observed non-permanent water | 0.91 | 0.77–1.17 | 3,810 | 76 | 0.74, 1.35 |
| *Capped table (10,000 cells per location group)* | | | | | |
| EMS labels | 1.01 | 0.77–1.31 | 361 | 52 | — |
| EMS labels, flooded fraction over observed non-permanent water | 1.02 | 0.78–1.32 | 372 | 54 | 0.68, 1.46 |

Table A50: The floodplain reading (riverine and wetland classes) within 500 m of settlements, given the terrain prediction (exploratory, amendment 10, item B2; defined after the class estimates had been seen). Intervals: pairs cluster bootstrap over location groups, 2,000 replicates. Countries are those of the nearest settlement. $^{x}$Forms removed in the re-reading. Columns 4–5: flooded cells and location groups with such cells (or cells, for the forms).

| Estimate | Odds ratio | 95 % interval | Flooded | Groups |
|---|---|---|---|---|
| All AOIs (floodplain reading) | 1.45 | 0.93–2.08 | 163 | 85 |
| re-read through the same code, all forms | 1.42 | 0.90–2.02 | 163 | 91 |
| re-read without bank, au, dike and polder forms | 1.54 | 0.94–2.28 | 142 | 85 |
| Without activation EMSR097 | 1.39 | 0.75–2.28 | 82 | 83 |
| *One country left out* | | | | |
| without BD (81 flooded cells) | 1.39 | 0.75–2.23 | 82 | 83 |
| without DE (22 flooded cells) | 1.55 | 0.92–2.31 | 141 | 79 |
| without AU (16 flooded cells) | 1.35 | 0.83–1.84 | 147 | 77 |
| without MX (9 flooded cells) | 1.38 | 0.86–2.00 | 154 | 83 |
| without PK (8 flooded cells) | 1.55 | 0.98–2.25 | 155 | 80 |
| without SI (8 flooded cells) | 1.40 | 0.87–1.99 | 155 | 83 |
| without ES (6 flooded cells) | 1.40 | 0.86–2.03 | 157 | 80 |
| without GB (3 flooded cells) | 1.46 | 0.90–2.10 | 160 | 77 |
| without RO (3 flooded cells) | 1.44 | 0.88–2.10 | 160 | 82 |
| without BE (2 flooded cells) | 1.44 | 0.91–2.07 | 161 | 85 |
| without KH (2 flooded cells) | 1.45 | 0.92–2.08 | 161 | 85 |
| without AL (1 flooded cell) | 1.44 | 0.90–2.05 | 162 | 84 |
| without MM (1 flooded cell) | 1.45 | 0.93–2.05 | 162 | 84 |
| without VN (1 flooded cell) | 1.48 | 0.93–2.12 | 162 | 79 |
| *Composition by matched form (flooded cells within 500 m; top 12)* | | | | |
| riverine *char* | | | 71 | 340 cells |
| riverine *au*$^{x}$ | | | 19 | 459 cells |
| riverine *island* | | | 14 | 27 cells |
| wet *bil* | | | 9 | 59 cells |
| wet *estero* | | | 9 | 18 cells |
| riverine *otok* | | | 8 | 34 cells |
| riverine *arenal* | | | 6 | 19 cells |
| wet *kallar* | | | 5 | 7 cells |
| riverine *lunca* | | | 3 | 74 cells |
| riverine *bela* | | | 3 | 23 cells |
| riverine *kaoh* | | | 2 | 10 cells |
| riverine *flat* | | | 2 | 49 cells |

Table A51: Calibration of the held-out predictions (exploratory, amendment 8): intercept and slope of a logistic regression of the labels on the logit of the prediction, Brier score with the reliability and resolution terms of the decomposition of Murphy (1973) (ten equal-count bins), and median within-AOI area under the ROC curve. M1: baseline model; M1s: recalibrated; M1c: with the regional composition of names (reference of the primary comparison); M3n: with the names. Skill: each AOI predicted flooded on as many cells as the labels flood (CSI: critical success index). The primary, country, time and language-model splits use the confirmatory run (frozen code); the Brier scores of the corrected run, which differ in the fourth decimal, are in Table A23.

| Split | Model | Intercept | Slope | Brier | Reliability | Resolution | AUC |
|---|---|---|---|---|---|---|---|
| Primary | M1 | -0.27 | 0.85 | 0.0437 | $3.4 \times 10^{-4}$ | $7.7 \times 10^{-3}$ | 0.90 |
| | M1s | 0.09 | 1.05 | 0.0434 | $5.9 \times 10^{-5}$ | $7.7 \times 10^{-3}$ | 0.90 |
| | M1c | -0.23 | 0.93 | 0.0431 | $2.1 \times 10^{-4}$ | $8.0 \times 10^{-3}$ | 0.90 |
| | M3n | -0.23 | 0.93 | 0.0431 | $2.2 \times 10^{-4}$ | $8.0 \times 10^{-3}$ | 0.90 |
| Countries held out | M1 | -0.57 | 0.69 | 0.0494 | $1.6 \times 10^{-3}$ | $5.5 \times 10^{-3}$ | 0.90 |
| | M1s | -0.18 | 0.95 | 0.0479 | $6.7 \times 10^{-4}$ | $5.4 \times 10^{-3}$ | 0.90 |
| | M1c | -0.39 | 0.84 | 0.0485 | $8.1 \times 10^{-4}$ | $5.5 \times 10^{-3}$ | 0.90 |
| | M3n | -0.39 | 0.84 | 0.0485 | $7.9 \times 10^{-4}$ | $5.5 \times 10^{-3}$ | 0.90 |
| Floods from 2024 | M1 | -0.88 | 0.75 | 0.0472 | $2.9 \times 10^{-3}$ | $3.4 \times 10^{-3}$ | 0.90 |
| | M1s | -0.58 | 0.95 | 0.0460 | $2.0 \times 10^{-3}$ | $3.4 \times 10^{-3}$ | 0.90 |
| | M1c | -0.70 | 0.91 | 0.0468 | $2.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | 0.90 |
| | M3n | -0.71 | 0.91 | 0.0468 | $2.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | 0.90 |
| After the LM cut-off | M1 | -0.69 | 0.80 | 0.0499 | $1.9 \times 10^{-3}$ | $4.2 \times 10^{-3}$ | 0.90 |
| | M1s | -0.44 | 0.97 | 0.0494 | $1.5 \times 10^{-3}$ | $4.2 \times 10^{-3}$ | 0.90 |
| | M1c | -0.53 | 0.97 | 0.0497 | $2.0 \times 10^{-3}$ | $4.3 \times 10^{-3}$ | 0.90 |
| | M3n | -0.54 | 0.97 | 0.0498 | $2.0 \times 10^{-3}$ | $4.3 \times 10^{-3}$ | 0.91 |
| Events held out whole, frozen code | M1 | -0.56 | 0.70 | 0.0488 | $1.4 \times 10^{-3}$ | $5.3 \times 10^{-3}$ | 0.90 |
| | M1s | -0.13 | 0.96 | 0.0477 | $3.7 \times 10^{-4}$ | $5.1 \times 10^{-3}$ | 0.90 |
| | M1c | -0.69 | 0.74 | 0.0518 | $2.1 \times 10^{-3}$ | $5.4 \times 10^{-3}$ | 0.90 |
| | M3n | -0.68 | 0.74 | 0.0516 | $2.1 \times 10^{-3}$ | $5.4 \times 10^{-3}$ | 0.90 |
| Events held out whole, corrected code | M1 | -0.56 | 0.70 | 0.0488 | $1.4 \times 10^{-3}$ | $5.3 \times 10^{-3}$ | 0.90 |
| | M1s | -0.13 | 0.96 | 0.0477 | $3.7 \times 10^{-4}$ | $5.1 \times 10^{-3}$ | 0.90 |
| | M1c | -0.70 | 0.74 | 0.0519 | $2.2 \times 10^{-3}$ | $5.4 \times 10^{-3}$ | 0.90 |
| | M3n | -0.70 | 0.74 | 0.0518 | $2.2 \times 10^{-3}$ | $5.4 \times 10^{-3}$ | 0.90 |
| GFM labels, refitted | M1 | -1.13 | 0.38 | 0.0858 | $6.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | 0.86 |
| | M1s | -0.48 | 0.79 | 0.0778 | $8.2 \times 10^{-4}$ | $2.9 \times 10^{-3}$ | 0.86 |
| | M1c | -1.99 | 0.16 | 0.1155 | $2.5 \times 10^{-2}$ | $2.2 \times 10^{-3}$ | 0.86 |
| | M3n | -1.99 | 0.16 | 0.1150 | $2.5 \times 10^{-2}$ | $2.2 \times 10^{-3}$ | 0.85 |

*Skill in flood-mapping terms, primary split, median over AOIs (interquartile range)*

| | |
|---|---|
| Baseline model (M1) | CSI 0.13 (0.03–0.27); hit rate 0.23 (0.07–0.42); false alarms 0.77 (0.58–0.93); AUC 0.90 (0.80–0.96); 479 AOIs |
| HAND alone (mean HAND of the cell; corrected run) | CSI 0.08 (0.00–0.21); hit rate 0.16; false alarms 0.84; 479 AOIs |

# How the names were read

Table A52: Checks of the lexicon and of the readings (exploratory, amendment 13, items B10 to B12 and C2). Counts are of the settlements nearest to at least one evaluable cell of the uncapped table, read with the frozen lexicon. Forms are matched in the ASCII field of GeoNames; Unicode names are read with umlauts, å, ø folded to one letter, æ to ae and ß to ss. Both coders' codes are in `docs/verification/v70_two_coders.csv` of the code repository.

| Quantity | Value |
|---|---|
| *Analysed settlements matched by homographs and wrong-language readings* | |
| German *-ach* not in *-bach* | 57 |
| German *-ach* in *-bach* | 427 |
| Norwegian *-ore*, *-ora* | 2 |
| of them in *Store-*, *Stora-* | 0 |
| French *mont-* as a prefix | 228 |
| of them *Montreuil*, *Montier*, *Moutier* | 12 |
| German *Ried* | 75 |
| Punjabi/Urdu *kacha* | 1 |
| Burmese *taung* | 139 |
| Belgian *-eau* read as German *-au* | 9 |
| Italian names matched by the German block | 5 |
| Iran, *ab* as a word | 45 |
| Iran, *-ab* as an ending | 48 |
| Finnish *Ala* | 1 |
| Finnish *Yli*, *Ylä* | 1 |
| *Edited lexicon (amendment 13, item C2): entries removed or added (form × position)* | |
| Two-letter affixes and English transfer endings | 15 |
| Springs and wells | 145 |
| Rapids and waterfalls | 10 |
| Homographs | 15 |
| Relative qualifiers | 181 |
| Riverine entries named as ordinary meadows, islands or sand | 37 |
| Added (named omissions) | 13 |
| Entries before / after | 2,538 / 2,148 |
| *Water-related readings gained / lost when the Unicode name is read* | |
| bg | 0 / 1 |
| de | 4 / 0 |
| fi | 4 / 0 |
| km | 0 / 57 |
| pl | 0 / 2 |
| sv | 17 / 0 |
| tr | 0 / 3 |
| Agreement of the two coders on water-related derivations ($\kappa$) | 0.91 |
| Codes deposited (names in an audit sample left out) | 360 of 462 |
| *Language-model reading by lexicon block: settlements, water-related, unknown* | |
| fr | 13,745, 15 %, 2.0 % |
| km | 10,702, 39 %, 1.1 % |
| my | 9,547, 69 %, 1.8 % |
| de | 9,463, 21 %, 0.6 % |
| en | 8,963, 20 %, 2.4 % |
| it+de | 6,552, 16 %, 1.3 % |
| es+nah | 5,569, 13 %, 0.8 % |
| fa+hi | 5,332, 9.9 %, 4.1 % |

Table A53: The multilingual lexicon by block: languages covered, countries mapped to the block, number of distinct forms per class (water, wetland, riverine, crossing, lowland, highland), match positions used and number of spring and well forms removed in the strict reading. Countries are mapped to the blocks of their main languages; regional languages are not read.

| Block | Languages | Countries | Wa | We | Ri | Cr | Lo | Hi | Positions | Spr |
|---|---|---|---|---|---|---|---|---|---|---|
| en | English | AU BW CA CM ET FJ GB GH GY IE IM KE LK LR LS MT MW NA NG NZ PG PH SG SL SS SZ TZ UG US ZA ZM ZW | 25 | 16 | 16 | 9 | 18 | 25 | word prefix suffix | 6 |
| de | German | AT BE CH DE IT LI LU | 11 | 12 | 11 | 12 | 8 | 15 | word prefix suffix | 5 |
| nl | Dutch | BE NL SR | 9 | 7 | 11 | 5 | 4 | 6 | word prefix suffix | 0 |
| fr | French | BE BF BI BJ CA CD CF CG CH CI CM DJ DZ FR GN HT LU MA MC MG ML NE RW SN TD TG TN | 15 | 16 | 15 | 8 | 12 | 18 | word prefix suffix | 5 |
| it | Italian | CH IT SM VA | 19 | 16 | 14 | 6 | 13 | 16 | word prefix | 6 |
| es | Spanish | AD AR BO CL CO CR CU DO EC ES GT HN MX NI PA PE PR PY SV UY VE | 16 | 22 | 16 | 6 | 13 | 22 | word | 6 |
| pt | Portuguese | AO BR CV GW MZ PT TL | 18 | 17 | 12 | 7 | 8 | 14 | word | 5 |

| Block | Languages | Countries | Wa | We | Ri | Cr | Lo | Hi | Positions | Spr |
|---|---|---|---|---|---|---|---|---|---|---|
| el | Greek (romanized) | CY GR | 24 | 8 | 3 | 5 | 5 | 12 | word prefix | 11 |
| sh | Serbo-Croatian / Bosnian (Latin) | BA HR ME RS XK | 14 | 11 | 7 | 6 | 7 | 14 | word | 7 |
| sl | Slovene | SI | 7 | 6 | 5 | 2 | 5 | 12 | word | 3 |
| bg | Bulgarian / Macedonian (romanized) | BG MK | 9 | 6 | 3 | 2 | 7 | 13 | word | 6 |
| sq | Albanian | AL MK XK | 13 | 8 | 4 | 5 | 7 | 11 | word | 4 |
| ro | Romanian | MD RO | 17 | 10 | 10 | 8 | 9 | 14 | word | 5 |
| hu | Hungarian | HU | 8 | 6 | 4 | 4 | 4 | 7 | word prefix suffix | 2 |
| cs | Czech / Slovak | CZ SK | 15 | 13 | 8 | 6 | 8 | 15 | word | 5 |
| pl | Polish | PL | 13 | 15 | 12 | 5 | 6 | 14 | word | 4 |
| uk | East Slavic (Ukrainian / Russian / Belarusian, romanized) | BY GE KG KZ MD RU TJ TM UA UZ | 15 | 12 | 15 | 8 | 14 | 19 | word | 6 |
| lt | Lithuanian | LT | 9 | 6 | 6 | 4 | 5 | 11 | word suffix | 2 |
| lv | Latvian | LV | 7 | 4 | 6 | 3 | 4 | 5 | word suffix | 2 |
| et | Estonian | EE | 7 | 5 | 4 | 3 | 3 | 5 | word suffix | 1 |
| fi | Finnish | FI | 10 | 8 | 4 | 3 | 4 | 10 | word suffix | 1 |
| sv | Swedish | FI SE | 10 | 7 | 9 | 6 | 6 | 16 | word suffix | 2 |
| no | Norwegian (also used for Icelandic placeholders) | IS NO | 16 | 8 | 14 | 6 | 6 | 20 | word suffix | 1 |
| da | Danish | DK | 6 | 4 | 5 | 5 | 4 | 8 | word suffix | 1 |
| tr | Turkish | CY TR | 11 | 5 | 3 | 3 | 6 | 10 | word prefix suffix | 4 |
| ar | Arabic (romanized; Levant, Gulf, Maghreb conventions) | AE DJ DZ EG ER IL IQ JO LB LY MA OM PS SA SD SO SY TD TN YE | 19 | 12 | 8 | 6 | 13 | 16 | word | 8 |
| fa | Persian / Urdu / Pashto / Sindhi (romanized) | AF IR PK TJ | 26 | 8 | 11 | 7 | 14 | 16 | word prefix suffix | 4 |
| hi | Hindi / Bengali / Nepali (romanized) | BD IN NP PK | 17 | 8 | 11 | 4 | 8 | 15 | word suffix | 0 |
| vi | Vietnamese (asciiname; diacritics stripped — only low-collision syllables) | VN | 7 | 3 | 2 | 3 | 1 | 2 | word | 1 |
| id | Indonesian / Malay | ID MY SG TL | 13 | 9 | 7 | 7 | 6 | 8 | word prefix | 1 |
| tl | Filipino / Tagalog and related | PH | 7 | 4 | 5 | 4 | 4 | 7 | word | 1 |
| sw | Swahili | BI CD KE RW TZ UG | 7 | 5 | 2 | 4 | 2 | 6 | word | 2 |
| bantu | Southern/Central Bantu (Chichewa/Nyanja, Shona, Sena, Zulu/Xhosa, Sotho; broad set) | AO BW CD LS MW MZ SZ ZA ZM ZW | 15 | 4 | 2 | 4 | 3 | 10 | word prefix | 1 |
| af | Afrikaans | NA ZA ZW | 6 | 5 | 3 | 3 | 5 | 7 | word suffix | 2 |
| mg | Malagasy | MG | 6 | 4 | 3 | 2 | 4 | 10 | word prefix | 1 |
| qu | Quechua / Aymara (Andes) | BO EC PE | 11 | 2 | 0 | 0 | 2 | 9 | word suffix | 2 |
| nah | Nahuatl (Mexico) | MX | 3 | 0 | 0 | 0 | 0 | 3 | word suffix | 1 |
| my | Burmese (romanized) | MM | 4 | 0 | 1 | 0 | 0 | 3 | word | 0 |
| th | Thai / Lao (romanized) | LA TH | 7 | 4 | 2 | 2 | 3 | 7 | word | 0 |
| km | Khmer (romanized) | KH | 9 | 0 | 4 | 3 | 0 | 4 | word | 0 |
| zh | Chinese (pinyin; only low-ambiguity syllables) | CN TW | 2 | 0 | 0 | 1 | 0 | 1 | word suffix | 0 |
| ka | Georgian (romanized) | GE | 5 | 1 | 0 | 1 | 3 | 4 | word suffix | 2 |
| kk | Kazakh / Kyrgyz / Uzbek / Turkmen / Mongolian (romanized; Turkic core) | KG KZ MN TM UZ | 12 | 5 | 5 | 5 | 5 | 9 | word suffix | 2 |
| so | Somali | DJ SO | 7 | 3 | 1 | 2 | 3 | 4 | word | 1 |
| am | Amharic / Tigrinya (romanized) | ER ET | 7 | 2 | 0 | 2 | 4 | 5 | word | 2 |
| ng | Hausa / Yoruba / Igbo (Nigeria, Niger) | NE NG | 8 | 3 | 0 | 2 | 1 | 6 | word prefix | 0 |
| tupi | Tupi-derived Brazilian toponymy (prefixes only, high-precision) | BR | 7 | 0 | 0 | 0 | 0 | 3 | word prefix | 0 |

Table A54: The 20 water-related forms that matched the nearest name of the most analysed cells (amendment 6): language block, form, position, class, number of cells and of settlements. Overlapping forms are each counted: every name in *-bach* also matches *-ach*.

| Block | Form | Position | Class | Cells | Settlements |
|---|---|---|---|---|---|
| en | *creek* | word | water | 4,979 | 62 |
| fa | *chah* | word | water | 4,665 | 110 |
| en | *bridge* | word | crossing | 4,605 | 142 |
| en | *lake* | word | water | 3,002 | 35 |
| de | *ach* | suffix | water | 2,891 | 404 |
| de | *au* | suffix | riverine | 2,701 | 182 |
| de | *bach* | suffix | water | 2,541 | 360 |
| en | *dale* | suffix | lowland | 2,515 | 56 |
| hi | *char* | word | riverine | 2,303 | 149 |
| en | *ford* | suffix | crossing | 2,161 | 74 |
| en | *lower* | word | lowland | 1,679 | 39 |
| de | *nieder* | prefix | lowland | 1,260 | 99 |
| en | *vale* | word | lowland | 1,234 | 15 |
| ro | *lunca* | word | riverine | 1,211 | 7 |
| en | *flat* | word | riverine | 1,092 | 7 |
| sv | *dalen* | suffix | lowland | 1,067 | 11 |
| en | *burn* | suffix | water | 1,038 | 30 |
| en | *well* | suffix | water | 996 | 47 |
| pt | *vale* | word | lowland | 881 | 95 |
| no | *foss* | suffix | water | 820 | 8 |

Table A55: Audit of matched forms (exploratory, amendment 8): forms that the lexicon's own rules excluded (two-letter affixes), English suffixes that also read transferred and personal names, and regional words, with the number of analysed settlements (94,524 in all) whose name they match, the cells nearest those settlements, the countries and examples (most cells first).

| Block | Form | Settlements | Cells | Countries | Examples |
|---|---|---|---|---|---|
| de | *au* (suffix) | 182 | 2,701 | DE 162, AT 12, BE 8 | Choerau, Sandau, Obslau, Trebbichau |
| sv | *as* (suffix) | 158 | 3,282 | SE 156, FI 2 | Fryksas, Neistenkangas, Svartnas, Marnas |
| no | *as* (suffix) | 3 | 127 | NO 3 | Kjonnas, Raipas, Gevingas |
| fa | *ab* (suffix) | 73 | 589 | PK 40, IR 33 | Arab, Goth Qutab Brahui, Haji Tayyab, Haji Gulab |
| en | *burn* (suffix) | 30 | 1,038 | GB 22, US 6, AU 2 | Heyburn, Woodburn, Wythburn, Blackburn |
| en | *bourne* (suffix) | 5 | 99 | GB 4, IE 1 | Cranbourne, Pangbourne, Barbourne, Nailsbourne |
| en | *borne* (suffix) | 1 | 30 | AU 1 | Tichborne |
| en | *pool* (suffix) | 5 | 107 | GB 4, US 1 | Glenpool, Sampool, Bathpool, Firepool |
| en | *well* (suffix) | 47 | 996 | GB 26, US 14, IE 6, AU 1 | Brauncewell, Howell, Greenwell Point, Cranwell |
| en | *wells* (suffix) | 1 | 138 | GB 1 | Kingswells |
| hi | *char* (word) | 149 | 2,303 | BD 149 | Char Kodalia, Char Gobindapur, Char Gazi Para, Sonatani Char |
| fa | *bela* (word) | 9 | 200 | PK 9 | Bela Ali Khanana, Kot Bela, Bela Bhagwanpura, Makki Bela |
| fa | *kacha* (word) | 1 | 11 | PK 1 | Kacha Pakka Mor |
| fa | *pattan* (word) | 3 | 133 | PK 3 | Pattan Khurd, Bahkan Pattan, Sahuka Pattan |
| fa | *pattan* (prefix) | 1 | 10 | PK 1 | Chah Pattanwala |
| fa | *tibba* (word) | 9 | 79 | PK 9 | Tibba, Tibba Ghhale, Tibba Hussain Shah, Gharianwala Tibba |

*Most frequent matched forms by country (class; settlements / cells)*

| | |
|---|---|
| AU | *creek* (water; 33/4,374); *lake* (water; 9/2,573); *hill* (high; 30/2,442); *bridge* (crossing; 8/1,468); *dale* (low; 14/1,377) |
| DE | *ach* (water; 392/2,809); *berg* (high; 417/2,715); *au* (riverine; 162/2,642); *bach* (water; 354/2,530); *ober* (high; 213/1,419) |
| GB | *hill* (high; 142/3,049); *bridge* (crossing; 47/1,900); *ford* (crossing; 50/1,219); *dale* (low; 28/997); *ridge* (high; 25/860) |
| SE | *as* (high; 158/3,186); *berg* (high; 70/2,198); *asen* (high; 19/1,876); *berget* (high; 16/1,409); *dalen* (low; 14/1,080) |
| US | *hill* (high; 45/694); *creek* (water; 28/604); *valley* (low; 22/565); *ford* (crossing; 29/519); *ridge* (high; 31/443) |
| IT | *monte* (high; 65/1,277); *ponte* (crossing; 80/755); *piana* (low; 3/629); *poggio* (high; 21/561); *sotto* (low; 19/423) |
| FR | *mont* (high; 206/1,710); *haut* (high; 30/379); *pont* (crossing; 61/331); *haute* (high; 14/295); *motte* (high; 12/278) |
| PT | *monte* (high; 46/1,648); *vale* (low; 93/849); *cima* (high; 57/552); *agua* (water; 10/388); *porto* (crossing; 31/356) |
| NO | *foss* (water; 8/820); *berget* (high; 4/804); *berg* (high; 10/698); *dal* (low; 6/542); *botn* (low; 2/335) |
| RO | *lunca* (riverine; 7/1,211); *sus* (high; 9/818); *jos* (low; 10/744); *deal* (high; 7/715); *valea* (low; 9/503) |
| IR | *chah* (water; 35/3,421); *bala* (high; 15/396); *nahr* (water; 17/243); *koh* (high; 10/234); *ab* (water; 62/209) |
| BO | *loma* (high; 5/821); *pampa* (low; 11/531); *puentes* (crossing; 1/528); *pata* (high; 8/414); *baja* (low; 14/397) |
| ES | *arriba* (high; 2/285); *rio* (water; 12/284); *arroyo* (water; 3/255); *sierra* (high; 5/246); *cuesta* (high; 4/205) |
| MG | *ambohi* (high; 40/1,879); *ambato* (high; 14/628); *andrano* (water; 20/407); *vohi* (high; 4/261); *berge* (riverine; 1/7) |
| PE | *cerro* (high; 2/934); *alto* (high; 7/852); *cocha* (water; 3/336); *pampa* (low; 2/241); *loma* (high; 4/229) |

Table A56: Examples of names read as water-related by the lexicon, with the derivation recorded on Wikipedia (amendment 6): for each class, the two lowest-numbered coded names whose recorded derivation matches the class and the lowest-numbered one whose derivation does not. Quotations abridged; sources in the code repository.

| Class | Agreement | Name | Recorded derivation | Quotation |
|---|---|---|---|---|
| water | agrees | Windermere (GB) | water, other | "Windermere" is thought to translate as "Winand or Vinand's lake" ... 'mere', meaning 'lake' or 'pool' |
| water | agrees | Odolena Voda (CZ) | other, water | The name means "Odolen's water" ... named after Knight Odolen ... refers to the sufficient water in the area |
| water | disagrees | New Wells (US) | other | Austrian settlers referred to the community as Neu Wels, after the city in Austria, which was later Americanized to New Wells |
| wetland | agrees | Spoerkelnbruch (DE) | other, wet | Vermutlicher Namensgeber ist der Faulbaum ... Bruch ist eine alte Bezeichnung für ein Moor oder Feuchtgebiet |
| wetland | agrees | Clairmarais (FR) | wet | l'oïl clair + maresc : le « marais aux eaux claires » |
| wetland | disagrees | Spodnje Blato (SI) | low | Spodnje Blato (literally 'lower Blato') |
| riverine | agrees | L'Ile-Saint-Denis (FR) | riverine, other | L'Île-Saint-Denis doit son nom à l'abbaye de Saint-Denis ... qui eut seigneurie sur l'île centrale |
| riverine | agrees | Ostrvo (RS) | riverine | "Ostrovo" means "island" in Serbian. This name originates from the fact that Ostrovo is located on former Danube island |
| riverine | disagrees | Veliki Otok (SI) | water | Voda (reka Pivka) pa naj bi odtekala v Postojnsko jamo ... odtok se imenuje Veliki odtok ... dali vasi ime Veliki Otok |
| crossing | agrees | Pont-Pean (FR) | crossing, other | le toponyme Pont-Péan peut donc signifier « pont paysan » ou « pont construit ou possédé par un dénommé Péan » |
| crossing | agrees | Saint-Fargeau-Ponthierry (FR) | crossing, other | Saint-Fargeau est une déformation du nom de saint Ferréol ... Ponthierry trouverait son origine dans un pont sur l'École franchi par Thierry |
| crossing | disagrees | Ennerdale Bridge (GB) | low, other, water | The name Ennerdale means 'Anund's valley' ... cross- influenced by 'Ehen', the name of the local river |
| lowland | agrees | Dol pri Ljubljani (SI) | low | The name is derived from the Slovene common noun dol 'small valley' |
| lowland | agrees | Spodnji Velovlek (SI) | low | The name Spodnji Velovlek literally means 'lower Velovlek' |
| lowland | disagrees | Germigny-sous-Coulombs (FR) | other | Du latin germinare, « germer », du latin germinius, « rejet (botanique) » ou « rejeton, progéniture ». Coulombs fait référence à Coulombs-en-Valois |

Table A57: The readings in more detail (amendment 6). Precision against derivations recorded on Wikipedia by the position of the matched water-related form and by class (a name can match at more than one position, so the position rows overlap and their counts sum to more than the names); recall by class is unweighted and therefore not comparable with the weighted recall reported in the paper. Lower part: readings by continent and languages that are not read.

| Quantity | Count | Share |
|---|---|---|
| Precision, matched as a whole word | 51/71 | 72 % |
| Precision, matched as an ending | 25/30 | 83 % |
| Precision, matched as a beginning | 7/8 | 88 % |
| Class water: precision / recall (unweighted) | 31/38 / 31/57 | 82 % / 54 % |
| Class wetland: precision / recall (unweighted) | 3/6 / 3/15 | 50 % / 20 % |
| Class riverine: precision / recall (unweighted) | 8/11 / 8/13 | 73 % / 62 % |
| Class crossing: precision / recall (unweighted) | 16/18 / 16/21 | 89 % / 76 % |
| Class lowland: precision / recall (unweighted) | 18/37 / 18/34 | 49 % / 53 % |
| Class highland: precision / recall (unweighted) | 8/15 / 8/40 | 53 % / 20 % |
| Strict reading: precision / recall (weighted) | — | 83 % / 19 % |
| Water codes reclassified as springs or wells only | 9 | — |
| Cells whose nearest name is water-related, Africa: language model / lexicon | 144,265 cells | 44 % / 0.5 % |
| Cells whose nearest name is water-related, Asia: language model / lexicon | 196,090 cells | 31 % / 6.7 % |
| Cells whose nearest name is water-related, Europe: language model / lexicon | 824,854 cells | 37 % / 6.3 % |
| Cells whose nearest name is water-related, North America: language model / lexicon | 85,031 cells | 17 % / 6.8 % |
| Cells whose nearest name is water-related, Oceania: language model / lexicon | 150,806 cells | 53 % / 11 % |
| Cells whose nearest name is water-related, South America: language model / lexicon | 106,440 cells | 30 % / 3.1 % |
| Cells nearest a settlement in ES-Catalonia | 10,000 | 0.7 % |
| Cells nearest a settlement in ES-Basque Country | 174 | 0.0 % |
| Cells nearest a settlement in ES-Galicia | 1,528 | 0.1 % |
| Cells nearest a settlement in ES-Valencia | 20,346 | 1.3 % |
| Cells nearest a settlement in ES-Navarre | 1,113 | 0.1 % |
| Cells nearest a settlement in FR-Brittany | 8,229 | 0.5 % |
| Cells nearest a settlement in FR-Corsica | 9,905 | 0.7 % |
| Cells nearest a settlement in IE | 19,576 | 1.3 % |
| Cells nearest a settlement in AU | 145,844 | 9.7 % |
| Cells whose nearest name has a riverside construction | — | 0.3 % |

Table A58: Recorded etymologies on Wikipedia by lexicon class and by language of the name (post hoc, amendment 5). Upper part: among the names in each class of the lexicon whose Wikipedia section states a derivation, the share whose recorded derivation is in the same class and the share whose derivation is water-related in any class (the highland class is not water-related). Lower part, for languages with at least 30 coded sections: the share of water-related derivations among the names that the lexicon read as water-related (its precision) and among the other names, with counts. Shares are unweighted; intervals are 95 % (Wilson, 1927).

| Lexicon class | Coded | Same class | Water-related, any class |
|---|---|---|---|
| water | 38 | 82 % (67–91) | 84 % (70–93) |
| wetland | 6 | 50 % (19–81) | 100 % (61–100) |
| riverine | 11 | 73 % (43–90) | 82 % (52–95) |
| crossing | 18 | 89 % (67–97) | 94 % (74–99) |
| lowland | 37 | 49 % (33–64) | 54 % (38–69) |
| highland | 15 | 53 % (30–75) | 27 % (11–52) |
| **Language of the name** | **Coded** | **Water-related, read as water-related** | **Water-related, others** |
| English | 48 | 75 % (55–88); 18/24 | 29 % (15–49); 7/24 |
| French | 120 | 67 % (50–80); 22/33 | 18 % (12–28); 16/87 |
| Polish | 35 | 40 % (12–77); 2/5 | 3 % (1–17); 1/30 |
| Slovene | 54 | 94 % (73–99); 16/17 | 16 % (8–31); 6/37 |

Table A59: Classes that the language-model reading (N-read) assigned to the nearest names of the analysed cells (exploratory; amendment 10, item B15), counted once per name and weighted by the number of analysed cells nearest each name. “Unknown”: the model gave no class.

| Class | Names | Share | Cells | Share |
|---|---|---|---|---|
| water | 12,072 | 13 % | 234,593 | 16 % |
| wetland | 7,602 | 8.0 % | 125,945 | 8.4 % |
| riverine | 7,716 | 8.2 % | 105,543 | 7.0 % |
| crossing | 1,783 | 1.9 % | 29,676 | 2.0 % |
| lowland | 3,472 | 3.7 % | 61,728 | 4.1 % |
| highland | 8,832 | 9.3 % | 161,156 | 11 % |
| other environmental | 10,937 | 12 % | 148,830 | 9.9 % |
| non-environmental | 38,219 | 40 % | 550,235 | 37 % |
| unknown | 3,891 | 4.1 % | 89,780 | 6.0 % |

Table A60: The nearest settlements of the analysed cells by GeoNames feature code (exploratory; amendment 11, item B9): settlements, their share, those read as water-related and the cells nearest them. PPL: populated place; PPLA–PPLA4: seats of administrative divisions; PPLX: section of a populated place; PPLH: historical; PPLQ: abandoned; PPLW: destroyed; PPLL: locality; PPLF: farm village; PPLR: religious.

| Feature code | Settlements | Share | Water-related | Cells |
|---|---|---|---|---|
| PPL | 84,867 | 90 % | 5369 | 1,290,881 |
| PPLX | 3,907 | 4.1 % | 278 | 70,588 |
| PPLL | 1,807 | 1.9 % | 178 | 37,817 |
| PPLA3 | 1,669 | 1.8 % | 59 | 54,253 |
| PPLQ | 633 | 0.7 % | 33 | 7,390 |
| PPLA2 | 598 | 0.6 % | 26 | 25,149 |
| PPLA4 | 579 | 0.6 % | 84 | 12,093 |
| PPLF | 165 | 0.2 % | 16 | 1,726 |
| PPLA | 159 | 0.2 % | 9 | 5,234 |
| PPLH | 82 | 0.1 % | 3 | 1,780 |
| PPLCH | 20 | 0.0 % | 0 | 151 |
| PPLW | 18 | 0.0 % | 1 | 138 |
| PPLR | 11 | 0.0 % | 0 | 172 |
| PPLC | 4 | 0.0 % | 0 | 102 |
| PPLS | 4 | 0.0 % | 0 | 10 |
| PPLG | 1 | 0.0 % | 0 | 2 |
| Settlements whose coordinates are whole arc-minutes: 14 % | | | | |

Table A61: Settlements whose namesake is recorded on Wikidata (property P138), classed by the type of the namesake (amendment 5). The lexicon contains no river names by design, so it reads few settlements named after a river; most recorded namesakes are in France.

| Quantity | Count |
|---|---|
| Analysed settlements | 94,524 |
| with a Wikidata item | 26,509 |
| with a recorded namesake (P138) | 299 |
| named after a water feature | 139 |
| named after a person, event or other commemoration | 97 |
| Water-feature namesakes read as water-related by the lexicon | 9 of 139 (6 %) |
| Same, by the language model | 66 of 139 (47 %) |
| Lexicon water-related names with a water-feature namesake | 9 of 13 (69 %) |
| Language-model water-related names with a water-feature namesake | 66 of 92 (72 %) |
| Lexicon water-related names with a commemorative namesake | 2 of 13 (15 %) |

Table A62: Precision and recall of the lexicon's water-related reading by language of the article, for languages with at least 30 coded names (amendment 5; recall by language, amendment 7, addendum 1). Recall is estimated as a domain with the design weights of the whole sample (stratum size over sample size), whereas Table A63 weights the strata by their sizes within each country, with percentile intervals from 2,000 resamples within strata. The derivations were coded by the language model that compiled the lexicon. Strict: the strict reading (water without springs and wells, wetland, riverine) against strict derivations (amendment 8, addendum 7).

| Language | Coded names | Precision (95 % interval) | Strict precision | Recall (95 % interval) |
|---|---|---|---|---|
| fr | 120 | 22 of 33 (67 %; 50 %–80 %) | 5 of 7 (36 %–92 %) | 17 % (10 %–28 %) |
| sl | 54 | 16 of 17 (94 %; 73 %–99 %) | 8 of 9 (56 %–98 %) | 29 % (15 %–59 %) |
| en | 48 | 18 of 24 (75 %; 55 %–88 %) | 9 of 11 (52 %–95 %) | 28 % (16 %–56 %) |
| pl | 35 | 2 of 5 (40 %; 12 %–77 %) | 1 of 1 (21 %–100 %) | 23 % (0 %–100 %) |

Table A63: Precision and recall of the lexicon's water-related reading by country, against etymologies recorded on Wikipedia (exploratory, amendment 10, item B6): countries with at least ten coded names. Recall weights the two sampling strata by their sizes within the country, whereas Table A62 uses the design weights of the whole sample (stratum size over sample size); the two agree on precision and differ on recall where the sampling fraction differs between countries. Forms: the only matched forms with at least ten coded names read as water-related. Both coders are language models; coder 1 compiled the lexicon.

| Country | Coded | Read as water-related | Precision (1) | Recall (1) | Precision (2) | Recall (2) |
|---|---|---|---|---|---|---|
| FR | 120 | 33 | 67 % | 12 % | 61 % | 7.9 % |
| SI | 54 | 17 | 94 % | 21 % | 85 % | 19 % |
| PL | 35 | 5 | 40 % | 17 % | 80 % | 12 % |
| GB | 33 | 16 | 81 % | 20 % | 83 % | 23 % |
| CZ | 20 | 6 | 83 % | 33 % | 83 % | 20 % |
| MX | 10 | 8 | 88 % | 100 % | 88 % | 100 % |
| Form *sous* (coder 1) | | 12 | 25 % | | | |
| Form *sous* (coder 2) | | 11 | 27 % | | | |

Table A64: Analysed cells whose names are read in a language other than their own (exploratory, amendment 10, item B14). Country level: the first language of the GeoNames country record is not among the lexicon blocks applied to the country. Regions: Natural Earth first-level units (point in polygon of the cell centre).

| Country or region | Cells | Water-related | Flooded, 500 m | AOIs |
|---|---|---|---|---|
| *Countries whose first language is not among the lexicon blocks applied* | | | | |
| PK (ur; read as fa, hi) | 61,219 | 2,249 | 4534 | |
| BD (bn; read as hi) | 19,720 | 2,652 | 643 | |
| LK (si; read as en) | 9,032 | 0 | 147 | |
| MK (mk; read as bg, sq) | 5,132 | 256 | 0 | |
| TL (tet; read as id, pt) | 4,962 | 0 | 33 | |
| RU (ru; read as uk) | 2,070 | 0 | 0 | |
| SK (sk; read as cs) | 1,926 | 52 | 0 | |
| LA (lo; read as th) | 333 | 9 | 0 | |
| IN (en; read as hi) | 323 | 0 | 0 | |
| LU (lb; read as de, fr) | 6 | 0 | 0 | |
| *Regions with a language other than the one read* | | | | |
| Wales | 0 | 0 | 0 | 0 |
| Highland and Eilean Siar | 3,997 | 348 | 0 | 2 |
| Northern Ireland | 3,316 | 0 | 0 | 2 |
| Ireland | 18,290 | 2,220 | 21 | 19 |
| Catalan-speaking Spain | 30,255 | 93 | 58 | 5 |
| Basque Country and Navarre | 1,230 | 3 | 1 | 4 |
| Galicia | 1,528 | 0 | 0 | 1 |
| Brittany | 7,943 | 244 | 6 | 7 |
| Alsace | 0 | 0 | 0 | 0 |
| Friesland | 0 | 0 | 0 | 0 |
| Finnmark and Troms | 3,818 | 0 | 2 | 2 |
| Norrbotten | 2,621 | 0 | 11 | 4 |
| Lapland (Finland) | 11,124 | 1,124 | 11 | 5 |

# Checks after the eleventh critiques (amendment 15)

Table A65: Odds ratios of flooding that the margin and the upper limit of each interval of $R$ correspond to, and the direct estimates (exploratory; amendments 6 to 14). Each bound belongs to the interval named in its row; a bound is not a test inversion. Each column has one scale. Nominal: the odds ratio injected into the labels. Estimator: the same effect on the scale of the odds ratios of Eq. (2) of the paper, read from the estimator's values on labels with known injected effects (interpolated on the log scale; beyond the largest effect used for this conversion, an injected odds ratio of 3, written as greater than the estimator's value there). No-effect reference: the mean increment of reassigned names with the regional composition rebuilt (amendment 8); bounds with other references are in Tables S15, S16 and A32 of the Supplement. [a] Monte Carlo 2.5th to 97.5th percentiles over the simulated replicates (50 each), which reflect simulation error only. [b] Effect injected on a latent indicator of truly water-related names, drawn with the misreading rates measured against recorded etymologies. [c] Within 500 m, pairs cluster bootstrap over location groups (2,000 replicates); on all ground, cluster-robust interval. [d] Pairs cluster bootstrap (2,000 replicates). [e] Effects on all ground nearest a water-related name, with every model refitted on the injected labels and the decision table of the protocol; names were found useful at injected odds ratios of 1.25 in 0 of 40, 1.50 in 0 of 40 and 2.00 in 40 of 40 replicates (Table S17 of the Supplement).

| | All ground nearest the name | | Within 500 m of the settlement | |
|---|---|---|---|---|
| | Nominal | Estimator | Nominal | Estimator |
| *Injected effects (the baseline model's held-out prediction taken as the true probability; base models fixed unless the row says refitted)* | | | | |
| Odds ratio at the margin | 2.33 | 2.94 | > 20 | — |
| effect that differs between settlements (mean odds ratio) | 2.30 | — | — | — |
| *Odds ratio reached by the upper limit of each interval of R (limit in parentheses)* | | | | |
| Protocol's interval (0.003; Monte Carlo range)[a] | 1.25 (1.24–1.26) | 1.36 | 6.25 (6.15–6.36) | > 4.73 |
| Spread over fold assignments (amendment 5; 0.004) | 1.31 | 1.45 | 7.77 | > 4.73 |
| Location groups, second stage refitted (amendment 8; 0.013) | 1.57 | 1.83 | 18.37 | > 4.73 |
| Activations × countries, second stage refitted (amendment 11; 0.036) | 2.11 | 2.63 | > 20 | — |
| Test on the ratio (a): location groups, models fixed (amendment 13; 0.004) | 1.29 | 1.43 | 7.40 | > 4.73 |
| Test on the ratio (b): location groups, refitted (0.016) | 1.65 | 1.95 | > 20 | — |
| Test on the ratio (c): activations within countries, refitted (0.031) | 1.99 | 2.45 | > 20 | — |
| Test on the ratio (d): (c) with fold-assignment variance (0.031) | 1.99 | 2.45 | > 20 | — |
| for truly water-related names[b] | 1.40 | — | 11.01 | — |
| flips correlated within the ground of each settlement | 1.27 | 1.39 | — | — |
| effect that differs between settlements (mean odds ratio) | 1.23 | — | — | — |
| Lower bound for effects that remove flooding | 0.70 | — | < 0.05 | — |
| Detectable with 80 % power, normal approximation (fixed; refitted) | 1.56; 2.00 | — | 17.80; > 20 | — |
| Detected in at least 80 % of simulations with every model refitted[e] | 2.00 | — | — | — |
| *Direct estimates: water-related against other names, given the baseline model (95 % interval)* | | | | |
| EMS labels, capped sample[c] | — | 1.02 (0.83–1.25) | — | 1.01 (0.77–1.31) |
| EMS labels, every evaluable cell[d] | — | — | — | 0.91 (0.76–1.16) |
| GFM labels, every assessable cell[d] | — | — | — | 0.82 (0.60–1.07) |
| EMS labels on the same cells[d] | — | — | — | 0.89 (0.67–1.16) |

Table A66: The ground within 500 m of settlements with and without water-related names (exploratory, amendment 15, item B5; capped EMS table, frozen reading): mean shares and their difference within AOIs (AOI fixed effects, 95 % intervals clustered by location group). The rule fixed before computing: if every difference is below 0.05 in absolute value with its interval within ±0.05, the EMS odds ratios near settlements are described as bounds on the association, provided detection errors are the same around both groups. The rule was met.

| Quantity | Water-related | Other | Difference (95 %) | Cells |
|---|---|---|---|---|
| Built-up share (WorldCover 10 m) | 0.125 | 0.139 | −0.013 (−0.023 to −0.003) | 160,239 |
| Tree share (WorldCover 10 m) | 0.303 | 0.306 | −0.002 (−0.013 to 0.008) | 160,239 |
| Share of pixels GFM cannot assess (GFM AOIs) | 0.638 | 0.649 | −0.030 (−0.045 to −0.015) | 103,196 |
| Share of permanent water | 0.005 | 0.004 | 0.001 (−0.0005 to 0.002) | 160,239 |

Table A67: Inference for the odds ratios near settlements on the cells GFM can assess (exploratory, amendment 15, item B10): cluster-robust 95 % intervals with normal and $t(G-1)$ critical values, an unrestricted wild cluster score bootstrap over location groups (Webb weights, 9,999 draws), and the odds ratio over 200 permutations of the labels within AOIs with the names fixed (median, 2.5th–97.5th percentiles and the share of permutations below the observed odds ratio). The lower block gives the estimate of amendment 7 by thirds of flood magnitude on the capped EMS table, with cluster-robust intervals; AOIs whose cells are all flooded or all dry within the sample drop out of the estimate.

| Estimate | OR | CR, $z$ | CR, $t$ | Wild | Labels permuted | Groups |
|---|---|---|---|---|---|---|
| Planned GFM estimate | 0.55 | 0.37–0.82 | 0.36–0.83 | 0.37–0.81 | 0.99 (0.78–1.23; 0.0 %) | 80 |
| EMS labels on the same cells | 0.79 | 0.58–1.07 | 0.58–1.07 | 0.59–1.05 | 1.00 (0.78–1.22; 4.0 %) | 80 |
| *Capped EMS table by thirds of flood magnitude (amendment 7): flooded share of the AOI over its share of ground within 2 m of the drainage* | | | | | | |
| Lowest third (median magnitude 0.006; 167 AOIs, 7 flooded cells near water-related names) | 0.40 | 0.18–0.91 | | | | 83 |
| Middle third (median magnitude 0.026; 166 AOIs, 45 flooded cells near water-related names) | 0.99 | 0.68–1.45 | | | | 98 |
| Highest third (median magnitude 0.101; 167 AOIs, 309 flooded cells near water-related names) | 1.07 | 0.78–1.47 | | | | 94 |

Table A68: Census of the exploratory interval estimates of amendments 2 to 15 (amendment 15, item B8; unique, valid estimates; file v76_estimates.csv in the code repository), in three families: flood (estimates on the flood labels or the predictions), site (descriptions of the ground near settlements without labels) and nuisance (coefficients of distance bands and of the baseline prediction, and contrasts between distances): number, number excluding the no-effect value, and numbers with Benjamini–Hochberg (Benjamini and Hochberg, 1995) and Benjamini–Yekutieli (Benjamini and Yekutieli, 2001) $q$ below 0.05 within each family, with the direction of those excluding the no-effect value (favours or against the claim about names).

| Family | Estimates | Excluding null | BH $q < 0.05$ | BY $q < 0.05$ | Direction |
|---|---|---|---|---|---|
| flood | 866 | 176 | 74 | 57 | unclassified 66, against 58, favours 52 |
| nuisance | 146 | 126 | 126 | 114 | not applicable 126 |
| site | 190 | 98 | 79 | 44 | favours 66, unclassified 30, against 2 |
| All | 1,202 | 400 | 279 | 215 | |

# Checks after the twelfth critiques (amendment 16)

Table A69: One-sided 95 % upper limits of $R$ under every construction with saved resamples, with the rule for few clusters fixed in amendment 16 (item B3): $\nu$ is the Satterthwaite effective number of clusters of $L_+(\theta) = \Delta - \theta D$ evaluated at the upper limit $\theta_U$, iterated with the critical value $t_{0.95}(\nu-1)$; "no bound" when $\nu-1 < 1$ or the Fieller interval is unbounded. $G$: resampled units. The odds ratios are the injected odds ratios on all ground nearest a water-related name that each upper limit reaches, as a range over the four no-effect references of amendment 13 (item B4). Constructions whose limit with $t(\nu-1)$ exceeds the margin or is unbounded: 7.

| Construction | $R$ | $G$ | Upper $z$ | Upper $t(\nu)$ | Upper $t(\nu-1)$ | $\nu$ | OR ($z$) | OR ($t(\nu-1)$) |
|---|---|---|---|---|---|---|---|---|
| Protocol's run: location groups, models fixed | −0.003 | 195 | 0.003 | 0.006 | 0.007 | 4.3 | 1.20–1.50 | 1.31–1.60 |
| (a) Location groups, models fixed | −0.004 | 195 | 0.004 | 0.011 | no bound | 2.2 | 1.21–1.51 | no bound |
| (b) Location groups, second stage refitted | −0.004 | 195 | 0.016 | 0.026 | 0.034 | 3.4 | 1.58–1.82 | 2.01–2.22 |
| (c) Activations within countries, refitted | −0.004 | 55 | 0.031 | 0.064 | no bound | 2.9 | 1.93–2.16 | no bound |
| (d) As (c), with the variance of fold assignments | −0.004 | 55 | 0.031 | 0.065 | no bound | 2.9 | 1.93–2.16 | no bound |
| Location groups × activations (pigeonhole), refitted | −0.004 | 188 | 0.067 | no bound | no bound | 2.7 | 2.67–2.83 | no bound |
| Split keeping activations whole: super-clusters, refitted | 0.004 | 106 | 0.040 | 0.078 | no bound | 2.7 | 2.14–2.32 | no bound |
| the same, names centred within AOIs | −0.003 | 106 | 0.002 | 0.008 | no bound | 2.5 | 1.15–1.46 | no bound |
| Designated: super-clusters of the primary split, refitted | −0.003 | 106 | 0.014 | 0.030 | 0.141 | 2.5 | 1.53–1.78 | 3.97–4.18 |
| the same, names centred within AOIs | −0.004 | 106 | 0.001 | 0.003 | 0.004 | 4.6 | 1.14–1.44 | 1.20–1.50 |

Table A70: Where the names increment comes from, and checks of the readings (exploratory, amendment 16, item B9; frozen reading, the designated refit of item B1). Contribution: the group's or block's share of the pooled $\Delta = \mathrm{BS(M1c)} - \mathrm{BS(M3n)}$ (negative: names worsened the predictions); forms: the lexicon's water-related forms matched by the nearest names of the group's cells, with the number of cells. Blocks: the languages the lexicon assigns to the country of the nearest settlement.

| Group, block or subset | Share | Contribution | Cells | Forms |
|---|---|---|---|---|
| *Location groups with the largest shares of the variance of* $\Delta$ | | | | |
| g00175 (Pakistan) | 64 % | $-2.6 \times 10^{-5}$ | 10,000; 1.3 % flooded; radar | *chah* (word; 1071), *bela* (word; 85), *dara* (word; 39) |
| g00127 (Australia) | 13 % | $1.1 \times 10^{-5}$ | 4,190; 4.1 % flooded; radar | *meadow* (suffix; 257), *meadow* (word; 119), *lower* (word; 94) |
| g00025 (Bangladesh) | 3.0 % | $-5.7 \times 10^{-6}$ | 10,000; 23 % flooded; unknown | *char* (word; 871), *bil* (word; 100), *khola* (suffix; 36) |
| g00126 (Australia) | 2.3 % | $-5.0 \times 10^{-6}$ | 10,000; 28 % flooded; radar | *burn* (suffix; 179), *flat* (word; 130), *billabong* (word; 116) |
| g00007 (Germany) | 2.0 % | $4.3 \times 10^{-6}$ | 10,000; 22 % flooded; unknown | *au* (suffix; 678), *thal* (suffix; 263), *beck* (suffix; 98) |
| g00132 (Pakistan) | 1.7 % | $3.9 \times 10^{-6}$ | 10,000; 66 % flooded; optical | *ab* (suffix; 130), *darya* (word; 16), *kallar* (word; 16) |
| g00008 (Germany) | 1.7 % | $-4.2 \times 10^{-6}$ | 5,015; 11 % flooded; unknown | *au* (suffix; 638), *nieder* (prefix; 7) |
| g00191 (Portugal) | 1.4 % | $-3.9 \times 10^{-6}$ | 10,000; 6.8 % flooded; radar | *vale* (word; 534), *agua* (word; 365), *rio* (word; 233) |
| g00166 (Germany) | 1.0 % | $2.9 \times 10^{-6}$ | 10,000; 0.7 % flooded; radar | *thal* (suffix; 236), *au* (suffix; 231), *werder* (suffix; 139) |
| g00030 (Myanmar) | 0.9 % | $2.8 \times 10^{-6}$ | 10,000; 11 % flooded; unknown | *chaung* (word; 620), *kan* (word; 231), *kyun* (word; 99) |
| *Language blocks with the largest contributions to* $\Delta$ | | | | |
| fa+hi | — | $-2.3 \times 10^{-5}$ | 61,219 | — |
| hi | — | $-6.7 \times 10^{-6}$ | 20,043 | — |
| mg+fr | — | $-4.4 \times 10^{-6}$ | 23,568 | — |
| pt | — | $-4.3 \times 10^{-6}$ | 20,062 | — |
| es+qu | — | $-1.3 \times 10^{-6}$ | 76,440 | — |
| de | — | $-1.1 \times 10^{-6}$ | 83,889 | — |
| *Subsets of cells (models fixed; R and 90 % interval)* | | | | |
| Nearest settlement not on whole arc-minutes | — | −0.004 (−0.013 to 0.003) | 1,472,604 | centred −0.004 |
| Nearest settlement coded PPL | — | −0.004 (−0.014 to 0.004) | 1,290,881 | centred −0.005 |
| GeoNames record last modified before the event | — | −0.003 (−0.019 to 0.008) | 941,820 | centred −0.004 |
| The block reads the country's first language | — | 0.0003 (−0.004 to 0.006) | 1,308,823 | centred −0.002 |
| *Agreement of the two coders of recorded derivations (Cohen's* $\kappa$*)* | | | | |
| water 0.94 ; wet 0.97 ; riverine 0.79 ; crossing 0.97 ; low 0.89 ; high 0.90 ; wetside 0.91 | — | — | 350 | — |

# References


Baugh C, Colonese J, D'Angelo C, et al (2024) Modelled flood inundation for different return period scenarios at the global scale, version 2.1.2. European Commission, Joint Research Centre (JRC) [data set], URL https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/CEMS-GLOFAS/flood_hazard/

Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B (Methodological) 57:289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Hawker L, Uhe P, Paulo L, et al (2022) A 30 m global map of elevation with forests and buildings removed. Environmental Research Letters 17:024016. https://doi.org/10.1088/1748-9326/ac4d4f

Holm S (1979) A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6:65–70

Kish L (1965) Survey Sampling. Wiley, New York

Murphy AH (1973) A new vector partition of the probability score. Journal of Applied Meteorology 12:595–600. https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2

Nadeau C, Bengio Y (2003) Inference for the generalization error. Machine Learning 52(3):239–281. https://doi.org/10.1023/A:1024068626366

Owen AB (2007) The pigeonhole bootstrap. The Annals of Applied Statistics 1(2):386–411. https://doi.org/10.1214/07-AOAS122

Pekel JF, Cottam A, Gorelick N, et al (2016) High-resolution mapping of global surface water and its long-term changes. Nature 540:418–422. https://doi.org/10.1038/nature20584

Wilson EB (1927) Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association 22:209–212. https://doi.org/10.1080/01621459.1927.10502953